

Air Pollution and Economic Activity: Evidence from Foot Traffic Patterns*

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Abstract

I investigate how air pollution affects economic activity. Using over 600 million phone-location-based foot traffic data points from SafeGraph, I conduct a large-scale analysis to examine the causal effect of air pollution on activity patterns across the US. Using changes in local wind direction as an instrumental variable (IV) for air pollution, I characterize the dynamic response to pollution exposure. I find that PM_{2.5} reduces economic activity in both the short and medium run, with effects persisting for up to two weeks before gradually recovering. The reductions are widespread across different economic sectors, with recreational activities experiencing the largest decline. The effect is more pronounced in higher-income counties and areas with larger shares of children, suggesting greater awareness among wealthier or more vulnerable populations.

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1 Introduction

Air pollution imposes significant costs on human well-being. Its negative impacts on morbidity and mortality are well-documented (Currie and Neidell, 2005; He et al., 2016; Deschenes et al., 2017; Deryugina et al., 2019). Air pollution can also influence daily behaviors, affecting decisions about where people go and what they do. However, the extent to which air pollution affects economic activities remains largely unexplored. Prior studies typically focus on specific types of activities, such as visits to zoos, national parks, or movie theaters (Neidell, 2009; Keiser et al., 2018; He et al., 2022). Yet, air pollution may have much broader effects, affecting other important activities such as shopping and restaurant dining.¹ These widespread behavioral changes can carry significant economic implications, as reductions in economic activity due to air pollution can lower overall economic output. Quantifying these broader effects of air pollution on daily activities is important for understanding its impact on human welfare and for designing optimal environmental policies.

In this paper, I conduct the first large-scale estimation of the causal effect of daily air pollution fluctuations on economic activity in the United States. Conventional datasets on daily activities often focus on specific regions or sectors (e.g., hiking, cycling, or zoo visits), making it difficult to assess the broader effects of air pollution on economic activities. I address this issue using foot traffic data from SafeGraph, which aggregates de-identified geospatial data from millions of US smartphones. My dataset combines over 600 million trips across a broad range of industries with satellite-based air pollution data covering the continental United States from 2018 to 2021. A primary identification concern when estimating the effect of air pollution is endogeneity. For instance, existing estimates may be biased due to reverse causality, as visitation-related traffic could increase local air pollution. To address this concern, I use changes in wind direction as an instrumental variable (IV) for air pollution to estimate its causal effect on daily activities.

I find that air pollution leads to statistically and economically significant reductions in daily activities. On average, a 1 $\mu\text{g}/\text{m}^3$ increase in PM2.5 concentration—approximately 10 percent of the sample mean—results in a 0.29% decrease in daily activity in the short run (contemporaneous), which corresponds to a nationwide reduction of 24.5 million trips annually.² In the medium run (over one month), the

¹In my sample, retail trade, accommodation and food services, and entertainment sectors are the three largest sectors, accounting for more than 60% of trips.

²This is calculated as $0.29\% \times 0.06985 \text{ visits per person per day} \times 331.9 \text{ million people} \times$

cumulative effect of a one-day air pollution exposure deepens over two weeks, reaching a 1.27% decline, before gradually recovering. By the end of one month, the point estimate is about two times larger than the contemporaneous effect. However, as the time window extends, the estimates become less precise due to wider standard errors.

This reduction is significant across many industries, indicating that the effect of air pollution on daily activities is more widespread than previously thought. Entertainment and recreation activities experience the most substantial declines (a 1 $\mu g/m^3$ increase in PM2.5 concentration results in a 0.47% decrease in recreational activities), likely due to the greater flexibility of these activities compared to other daily routines. Furthermore, the reduction in visits is generally more pronounced for outdoor facilities than for indoor ones. This is presumably because outdoor activities amplify the negative health effects of pollution due to increased respiration and exposure.

I also examine heterogeneity in responses to air pollution across different income levels and demographic groups. The reduction in daily activities is more pronounced in counties with higher incomes and larger proportions of children, suggesting that wealthier or more vulnerable populations may have greater awareness of elevated air pollution levels. I find that minority populations exhibit smaller behavioral responses to air pollution, which may partially explain the disproportionately greater negative health impacts of air pollution on Black individuals compared to White individuals (Alexander and Currie, 2017; Gillingham and Huang, 2021). While previous literature on environmental injustice has primarily focused on the unequal distribution of pollution (Banzhaf et al., 2019; Jbaily et al., 2022), my findings emphasize the role of avoidance behavior in exacerbating these disparities. Even when exposed to the same levels of air pollution, low-income and minority groups are less likely to adjust their behavior to mitigate exposure, which may further deepen environmental justice issues. Given that these groups are disproportionately likely to live in more polluted areas (Mikati et al., 2018; Heblich et al., 2021; Tessum et al., 2021), addressing this unequal burden may require targeted policy interventions.

This paper has three main contributions. First, it provides the first large-scale estimation of the causal effect of air pollution on daily activities in the United States. Existing literature studying the effects of air pollution on daily activities generally relies on limited samples from specific regions (Bresnahan et al., 1997; Zivin and Neidell, 2009), or a specific activity type, such as visits to national parks (Keiser

365 days $\approx$ 24.6 million trips annually.

et al., 2018), camping (Gellman et al., 2022) or movie watching (He et al., 2022), which makes the generalizability of these estimates unclear. In contrast, my analysis uses nationwide phone-location data and examines a broader range of activities, making it more representative than previous studies. My findings suggest that air pollution leads to reductions in activities across most industries, indicating that the behavioral response to air pollution is more widespread than previously understood.

Second, this paper reinforces the importance of characterizing avoidance behavior when quantifying the externalities of air pollution. The results show that individuals, especially vulnerable groups, actively engage in self-protective behaviors to reduce their exposure, such as spending more time at home. As a result, estimates that ignore these behaviors are likely biased downward, since health impacts would be greater without such avoidance actions. These avoidance responses also suggest that the marginal disutility of pollution exposure outweighs the marginal utility people derive from daily activities. Additionally, avoidance behavior itself can be costly, either through increased expenditures (Ito and Zhang, 2020; Zhang and Mu, 2018) or utility losses. For example, staying home and limiting daily activities can reduce physical activity, leading to health issues such as obesity (Hankinson et al., 2010), depression, and anxiety (Paluska and Schwenk, 2000), which impose further societal costs. Lastly, behavioral changes can negatively impact GDP by disrupting key industries. Given that avoidance behavior is widespread, accurately quantifying these effects is crucial for understanding the true costs of air pollution and determining optimal policies.

Third, this paper investigates whether individuals respond to day-to-day pollution fluctuations, adding to the relatively understudied topic of avoidance behavior. Previous studies show that air quality alerts prompt avoidance behavior (Neidell, 2009; Altindag et al., 2017). However, air quality warnings are rare and triggered only when the Air Quality Index exceeds a certain level, while negative effects of air pollution increase even before this threshold (Zivin and Neidell, 2009). Therefore, how individuals respond to these alerts does not necessarily reflect how they respond to air quality itself. In the absence of an air quality alert, individuals might not be aware of elevated pollution levels. My results are not solely driven by air quality alerts, indicating that people adjust their activities in response to more common, everyday pollution fluctuations. This finding suggests that avoidance behavior is more widespread than previously recognized, which in turn implies that the costs of pollution are underestimated.

The rest of the paper is organized as follows. Section 2 describes the data and

provides summary statistics. Section 3 presents the conceptual framework. Section 4 introduces the empirical strategy. Section 5 presents the main results, examines heterogeneity, explores mechanisms, and conducts a welfare analysis. Section 6 reports robustness checks. Section 7 concludes.

2 Data

The data used in the paper come from three main sources: mobile phone-based economic activity data from SafeGraph, satellite-based air pollution from Copernicus Atmosphere Monitoring Service global reanalysis (EAC4), and satellite-based weather data from EAC4 and European Centre for Medium-Range Weather Forecasts Reanalysis 5 (ERA5). The linkages and further details are described below.

2.1 Economic Activity Data

I obtain the economic activity data from SafeGraph.³ The dataset includes information collected from over 45 million smart mobile devices and covers over 3.6 million Points of Interest (POI) across the United States. In total, I obtain over 600 million trips from January 1, 2018, to December 30, 2021, across the United States.

SafeGraph conducts a data quality evaluation by comparing its demographic data with the American Community Survey (ACS) data from the US Census and found that their data are statistically representative of the population at the county level and above (Squire, 2019; Chang et al., 2022). Therefore, for empirical analysis, I match each location to its county based on latitude and longitude, and then aggregate visits at the county level. After aggregating all visits at the county level, I have 4,507,400 county-day observations. Additionally, since the dataset includes industry categories based on the North American Industry Classification System (NAICS) code, I am able to analyze foot traffic to different categories separately. The number of county-day observations for each category is shown in Table 1.⁴ As shown in Table A1, the vast majority of trips documented in the SafeGraph data are business-related, including visits to retail stores, hotels, restaurants, and entertainment facilities, which account for 57% of the total raw visits.

Because this dataset does not contain socioeconomic and demographic information about mobile device users for privacy protection reasons, I obtain county-level

³<https://www.safegraph.com/>. Accessed Sep 12, 2023.

⁴Note that the number of observations varies across categories because some counties may not have facilities of certain types.

population and income data from the United States Census Bureau. Since county populations vary widely and more populated counties tend to have more visits, I use visitation rates rather than raw visitation numbers as the dependent variable. To calculate county-level visitation rates, I aggregated the total number of visits in each county and then divided by the county’s total population.

2.2 Air Pollution Data

Although the United States Environmental Protection Agency (EPA) has been reporting air quality and other atmospheric data since 1970 and the number of pollutant monitors has increased over the years, there are still limitations: more than half of the monitors collect data on a 1-in-3 day schedule or 1-in-6-day schedule, resulting in a lack of data on certain days.⁵ Interpolating the missing data on these days can lead to bias, as air quality on unmonitored days tends to be worse than on monitored days due to strategic responses (Zou, 2021).

Therefore, rather than using monitor-based data, I use satellite-based air pollution data from the EAC4 reanalysis database.⁶ EAC4 reports PM2.5 and other atmospheric data every 3 hours with a $0.75^\circ \times 0.75^\circ$ grid ($\approx 81km \times 81km$), which is derived from a combination of satellite observations and atmospheric model simulations. I construct the county-level daily PM2.5 level in the following manner: for counties that have multiple satellite data, I average the gridded values overlapping each county; for counties that do not have satellite data, I interpolate their PM2.5 levels using inverse distance weighting (IDW) based on their latitude and longitude. Then, I match the visitation data with the air pollution data using county code and date. Figure A1 shows the average county-level visit rates and PM2.5 levels from January 1, 2018, to December 30, 2021. As a few counties do not have any visitation data for leisure facilities from SafeGraph, there are some missing values in the figure.

2.3 Weather Data

The analysis in this paper contains a flexible set of control variables for weather, including temperature, wind speed, and precipitation. Additionally, wind direction is used as the instrument for PM2.5 concentrations.

⁵See EPA’s Sampling Schedule Calendar: <https://www.epa.gov/amtic/sampling-schedule-calendar>. Accessed October 10, 2022.

⁶See <https://ads.atmosphere.copernicus.eu/cdsapp#!/dataset/cams-global-reanalysis-eac4?tab=overview>. Accessed September 19, 2022.

Daily temperature, wind direction, and wind speed data are also obtained from the EAC4 reanalysis database. I average the daily measures across all grid points in a particular county to obtain the county-level daily measure. For counties without satellite data, I interpolate their temperature, wind direction, and wind speed using IDW based on their latitude and longitude. Specifically, wind direction and wind speed are constructed using the East-West wind vector (u-wind) and the North-South wind vector (v-wind) provided in the database.⁷ Wind direction is defined as the direction the wind is blowing from.

In addition, I obtain precipitation data from the Copernicus ERA5 reanalysis hourly databases. Precipitation data are reported on a $0.25^\circ \times 0.25^\circ$ grid ($\approx 27km \times 27km$). I constructed the county-level daily precipitation by averaging the hourly data on a given day with grid points within a particular county. For counties without satellite data, I interpolate their precipitation using IDW based on their latitude and longitude.

2.4 Summary Statistics

Table 1 displays summary statistics for the main estimation sample, consisting of 4,507,400 county-day observations. The average daily concentration of PM2.5 is $11.5 \mu g/m^3$, with a standard deviation of 16.64.⁸ Demographic characteristics are measured at the county level in 2017. On average, counties in the sample have a population of approximately 105,050, with about 5.8% of residents under age 5, 77% White residents, and a per capita income of roughly \$26,000.

The average daily visit rate across all POIs within counties is 69.85 visits per 1,000 people.⁹ The Retail Trade sector exhibits the highest mean visit rate at 22.50 per 1,000 people, followed by the Accommodation and Food Services sector at 14.58 visits per 1,000 people. Most counties have at least some activity in each sector; a notable exception is the Mining, Quarrying, and Oil and Gas Extraction sector, where relevant POIs are missing in the majority of counties, resulting in very few

⁷Note that wind directions and speed are vectors, so they cannot be averaged or interpolated numerically. Therefore, when averaging or interpolating, I first take the average of the two vectors and then calculate the average wind direction and wind speed using the average vectors.

⁸This is slightly higher than the average PM2.5 concentration calculated using EPA ground monitors. One possible explanation for this discrepancy is strategic monitor placement, as discussed in Grainger et al. (2019).

⁹Note that the data from SafeGraph were collected from over 45 million mobile devices, roughly 14% of the US population. The raw number of visits before aggregating to the county level is reported in Table A1.

observations.

Table 1. Summary Statistics

Variables	Mean	SD	N
Pollution			
PM2.5 ($\mu g/m^3$)	11.50	16.64	4,507,400
Weather			
Temperature ($^{\circ}C$)	13.53	10.73	4,507,400
Total Precipitation (mm)	0.31	0.70	4,507,400
Wind Direction (degrees)	193.19	94.75	4,507,400
Wind Speed (m/s)	2.73	1.55	4,507,400
Dew Point ($^{\circ}C$)	6.81	10.81	4,507,400
Visibility (km)	17.55	3.87	4,507,400
Demographic			
Population	105,050	336,591	3,100
Age under 5 (%)	5.84	1.19	3,100
White (non-Hispanic) (%)	77.13	19.77	3,100
Per Capita Income (USD)	26,011	6,215	3,100
Visit Rates (per 1,000 people)			
All POIs	69.85	41.51	4,507,400
44-45: Retail Trade	22.50	14.28	4,506,558
72: Accommodation and Food Services	14.58	12.39	4,491,452
61: Educational Services	8.03	8.84	4,466,749
71: Arts, Entertainment, and Recreation	6.48	9.13	4,322,150
53: Real Estate and Rental and Leasing	5.73	7.44	3,858,106
62: Health Care and Social Assistance	4.88	3.86	4,449,158
81: Other Services	4.54	6.01	4,495,995
48-49: Transportation and Warehousing	1.33	2.62	4,482,408
92: Public Administration	1.11	1.55	4,468,576
31-33: Manufacturing	0.87	2.43	3,875,683
52: Finance and Insurance	0.73	0.72	4,282,607
51: Information	0.46	0.78	3,944,703
42: Wholesale Trade	0.45	0.76	3,677,317
54: Professional, Scientific, and Technical Services	0.41	0.48	4,040,757
23: Construction	0.31	0.50	3,614,912
55: Management of Companies and Enterprises	0.31	1.70	1,383,221
22: Utilities	0.24	0.63	2,228,604
56: Administrative and Support and Waste Services	0.18	0.34	3,009,650
21: Mining, Quarrying, and Oil and Gas Extraction	0.17	0.20	8,750
11: Agriculture, Forestry, Fishing and Hunting	0.11	0.27	658,854

Notes: Demographic variables are measured at the county level in 2017.

3 Conceptual Framework

To motivate the empirical analysis, I consider a simple dynamic setting in which individuals choose daily activity a_t to balance the enjoyment of activity against the

health costs of pollution exposure. Per-period utility is given by:

$$U_t = u(a_t, h_t) - r_t \phi(a_t),$$

where a_t represents activity level, h_t is the health stock, and r_t is ambient pollution. I assume $u_a(a_t, h_t) > 0$, $u_{aa}(a_t, h_t) < 0$, and $u_h(a_t, h_t) > 0$, reflecting diminishing marginal utility of activity and positive marginal utility of health. Pollution reduces utility via discomfort or symptoms experienced during activity, captured by $r_t \phi(a_t)$, where $\phi'(a_t) > 0$ and $\phi''(a_t) \geq 0$.

Health evolves according to partial adjustment:

$$h_{t+1} = h_t + \rho(\bar{h} - h_t) - \theta(r_t) a_t, \quad (1)$$

where $\rho \in (0, 1)$ is the recovery rate toward baseline health $\bar{h}$, and $\theta(r_t)$ is the pollution-induced damage per unit of activity. I assume $\theta'(r_t) > 0$, implying that higher pollution increases marginal harm.

The individual solves the intertemporal problem:

$$\max_{\{a_t\}_{t=0}^T} \sum_{t=0}^T \beta^t [u(a_t, h_t) - r_t \phi(a_t)] \quad \text{s.t.} \quad h_{t+1} = h_t + \rho(\bar{h} - h_t) - \theta(r_t) a_t,$$

where $\beta \in (0, 1]$ is the intertemporal discount factor. The Lagrangian is:

$$\mathcal{L} = \sum_{t=0}^T \beta^t [u(a_t, h_t) - r_t \phi(a_t)] + \sum_{t=0}^T \beta^{t+1} m_{t+1} [h_t + \rho(\bar{h} - h_t) - \theta(r_t) a_t - h_{t+1}],$$

where m_t is the shadow value of health at the start of period t . The first-order condition (FOC) with respect to a_t is:

$$u_a(a_t, h_t) - r_t \phi'(a_t) - \beta \theta(r_t) m_{t+1} = 0, \quad (2)$$

and m_t evolves as:

$$m_t = u_h(a_t, h_t) + \beta(1 - \rho) m_{t+1}. \quad (3)$$

Intuitively, in Equation (2), the marginal enjoyment of activity today $u_a(a_t, h_t)$ equals its marginal costs: the marginal disutility today $r_t \phi'(a_t)$ and the discounted shadow cost of tomorrow's health loss $\beta \theta(r_t) m_{t+1}$. Equation (3) defines the shadow value of health: one extra unit of health at the start of day t is worth its direct marginal

utility today, $u_h(a_t, h_t)$, plus the discounted continuation value that carries forward at rate $1 - \rho$.

Prediction 1 (Same-day avoidance). Define $F(a_t, r_t, m_{t+1}; h_t) \equiv u_a(a_t, h_t) - r_t \phi'(a_t) - \beta \theta(r_t) m_{t+1} = 0$. By the implicit function theorem,

$$\frac{\partial a_t^*}{\partial r_t} = -\frac{F_r}{F_a} = \frac{\phi'(a_t) + \beta \theta'(r_t) m_{t+1}}{u_{aa}(a_t, h_t) - r_t \phi''(a_t)} < 0,$$

since the numerator is positive and the denominator is negative. Thus, higher pollution reduces same-day activity.

Prediction 2 (Persistence and rebound). From Equation (1),

$$\frac{\partial h_{t+1}}{\partial r_t} = -\theta'(r_t) a_t - \theta(r_t) \frac{\partial a_t^*}{\partial r_t}.$$

The first term is negative, and the second term is positive (since $\partial a_t^*/\partial r_t < 0$), so overall $\partial h_{t+1}/\partial r_t < 0$ under reasonable conditions. That is, pollution reduces next-day health even accounting for behavioral avoidance. Lower h_{t+1} raises the shadow value m_{t+1} by Equation (3). If activity and health are complements ($u_{ah}(a, h) \geq 0$), the FOC at $t+1$ implies that a_{t+1} remains suppressed even after r_{t+1} normalizes. Iterating Equation (1) yields:

$$h_{t+k} - \bar{h} = (1 - \rho)^k (h_t - \bar{h}) - \sum_{j=0}^{k-1} (1 - \rho)^{k-1-j} \theta(r_{t+j}) a_{t+j},$$

so health converges toward $\bar{h}$ at rate ρ when pollution normalizes, leading to gradual recovery in activity. This persistence-and-rebound pattern is consistent with the empirical results shown in Figure 3.

Prediction 3 (Heterogeneity). To capture heterogeneity, let $\lambda(y)$ scale the exposure cost by socioeconomic characteristic y (e.g., income), with $\lambda'(y) > 0$. Rewriting Equation (2) with this term:

$$u_a(a_t, h_t) - \lambda(y) r_t \phi'(a_t) - \beta \theta(r_t) m_{t+1} = 0.$$

Differentiating with respect to y ,

$$\frac{\partial a_t^*}{\partial y} = \frac{-\lambda'(y) r_t \phi'(a_t)}{u_{aa}(a_t, h_t) - \lambda(y) r_t \phi''(a_t)} < 0.$$

Second, let biological vulnerability z (e.g., share of children) affect health damage: $\theta = \theta(r_t; z)$ with $\theta_z > 0$. Then:

$$\frac{\partial a_t^*}{\partial z} = \frac{\beta \theta_z(r_t; z) m_{t+1}}{u_{aa}(a_t, h_t) - r_t \phi''(a_t)} < 0.$$

Both channels imply larger activity reductions in groups with higher income (y) or higher child share (z), consistent with the empirical heterogeneity shown in Figure 5a and Figure 5b, respectively.

4 Empirical Strategy

4.1 Estimating Equations

My objective is to estimate the causal effect of acute (1-day) air pollution exposure on economic activity. I model this relationship using the following regression:

$$\frac{Y_{ct}^k}{Pop_c} = \beta^k \cdot PM2.5_{ct} + \mathbf{X}_{ct}^{k'} \gamma + \sigma_{cy} + \eta_{cm} + \mu_w + \psi_{my} + \epsilon_{ct} \quad (4)$$

where c indexes county, t indexes time at the daily level, y indexes year, and m indexes month. The outcome variable, Y_{ct}^k , represents the cumulative visit rate in county c over the k days following exposure on day t (including same-day visits). Visit rates are calculated as the total number of visits to all facilities in county c on date t , divided by the county's 2020 population. The parameter of interest, β^k , captures the effect of acute PM2.5 exposure on k -day economic activity. To ensure that β^k is not capturing the effect of weather conditions during the outcome window, the specification includes contemporaneous and k -lead values of the weather variables. I also include two lags of the weather controls to address potential confounding from past weather conditions. To minimize concerns about autocorrelation, my OLS estimates control for two leads and two lags of PM2.5, while my IV estimates control for two leads and two lags of the instruments.

My main specification controls for daily temperature, precipitation, wind speed, and dew point. I generate indicators for daily average temperatures falling into one of

14 bins, ranging from -9 to -6°C up to 24 to 27°C , with additional bins for temperatures below -9°C and above 27°C . I also generate indicators for precipitation quartiles, wind speed quintiles, and dew point quintiles.¹⁰ I then generate a set of indicators for all possible interactions of these weather variables and include them in my regressions as $\mathbf{X}_{ct}^k$. My estimates are robust to using less flexible weather controls or omitting weather controls entirely (Table 8). These results reinforce the assumption that my estimates are not driven by unobserved climatic factors that are correlated with both wind direction and economic activity.

In addition, I include a rich set of fixed effects, including county-by-year fixed effect σ_{cy} , county-by-month fixed effect η_{cm} , month-by-year fixed effect ψ_{my} and day-of-week fixed effect μ_w . Specifically, county-by-year fixed effects σ_{cy} pick up within-year variations in county-level factors that determine visits but are not captured by the control variables, such as demographic characteristics and economic conditions. County-by-month fixed effects η_{cm} control for seasonal unobservables across counties, such as different peak seasons due to different geographic features. Day-of-week fixed effect μ_w pick up cyclical visit patterns within the week. Lastly, month-by-year fixed effect ψ_{my} captures the time-varying shocks that are common in each month, such as economic recessions and pandemic outbreaks. I also examine the robustness of the results by including different fixed effects. The standard errors are clustered at the county level.

For OLS estimates of Equation (4) to be unbiased, the identifying assumption is that, conditional on control variables and fixed effects, unobserved determinants of visit rates (ϵ_{ct}) are uncorrelated with variation in PM2.5. Although high-frequency air pollution tends to be more random than long-term trends, some sources of endogeneity may still bias the estimate of β^k . First, there may be omitted variables that affect both ambient pollution and economic activity. For example, a local event could increase PM2.5 levels by raising traffic while simultaneously influencing individuals' time allocation. Second, although reverse causality is likely minimal, it cannot be entirely ruled out. If people choose to stay home rather than go out, the resulting decline in economic activity could also reduce local emissions.

To address this concern, I leverage the pollution variation due to changes in wind patterns to identify pollution impacts. Specifically, since wind directions are random, I use the changes in wind direction as an instrumental variable for air pollution

¹⁰Precipitation is categorized into quartiles instead of quintiles due to a high number of zero observations.

to derive the causal relationship (Deryugina et al., 2019). The assumption of this approach is that after controlling for covariates and fixed effects, changes in wind direction only affect people’s economic activity through their effects on air pollution. The specification for the first stage is:

$$PM2.5_{ct} = \sum_{g \in G} \sum_{b=0}^3 \pi_b^g \mathbf{1}[G_c = g] \cdot WindDir_{ct}^{90b} + \mathbf{X}_{ct}^{k'} \gamma + \sigma_{cy} + \eta_{cm} + \mu_w + \psi_{my} + \epsilon_{ct} \quad (5)$$

In Equation (5), the instrumental variable is constructed in the following manner. $WindDir_{ct}^{90b}$ equals 1 if wind direction in county c falls in the 90-degree interval $[90b, 90b+90)$ and 0 otherwise. Because the relationship between wind direction and pollution transport is location-specific, I allow the effect of each wind instrument on PM2.5, denoted by π_b^g , to vary across geographic regions.¹¹ To define these regions, I apply the K-means clustering algorithm to county centroids (latitude and longitude), classifying counties into 20 spatial groups. The clustering result is shown in Figure A2. $\mathbf{1}[G_c = g]$ equals 1 if county c is classified into monitor group g and 0 otherwise. Other control variables X_{ct} and fixed effects are defined as in Equation (4).

4.2 Identifying Variation

Air pollution in a given location can originate from sources that are either close by or far away. For instance, a city’s air quality may be affected by nearby traffic emissions, as well as by smoke from wildfires or pollutants released by power plants located hundreds of miles away. The source location matters because emissions from local sources tend to disperse unevenly across nearby areas, while emissions from more distant sources disperse more uniformly across the same region.

Equation (5) allows the effect of wind direction on air pollution to vary across geographic groups but not within a group. Intuitively, non-local sources located outside of the cluster are more likely to have similar effects on pollution levels in all (or most) counties in the cluster group. As a result, Equation (5) is more likely to capture variation in pollution driven by non-local sources. This is advantageous because pollution driven by local sources may not affect all individuals residing within the area in the same way, leading to measurement error.¹² In Section 6, I provide

¹¹Because the effect of wind direction on pollution depends on the surrounding geography, a westerly wind blowing clean ocean air into California is likely to have a very different impact on local pollution levels than a westerly wind carrying industrial emissions into a community located just east of Houston.

¹²Consider a local pollution source located in the center of a cluster. When the wind blows from

evidence that the pollution variation exploited in my design is primarily driven by non-local sources. Therefore, the effects of wind direction on pollution should be similar for all counties within the same geographic cluster. I employ 4 bins and 20 clusters for computational ease, and the results are robust to varying the number of wind direction bins and geographic clusters (Table A8).

Figure 1 illustrates my first-stage variation, using California as an example. The plots show the group-specific relationships between daily average wind direction and $\text{PM}_{2.5}$. Each specification controls for county-by-month, county-by-year, day-of-week, and month-by-year fixed effects, as well as flexible weather controls, in the same way as in my main specification. To show the wind–pollution relationship flexibly, I group wind direction into 30-degree bins, which necessitates estimating this modified first-stage equation separately for each geographic group. As shown in Figure 1, pollution levels are highest when the wind blows from the east-southeast direction, and lowest when the wind blows from the west, where the Pacific Ocean lies.

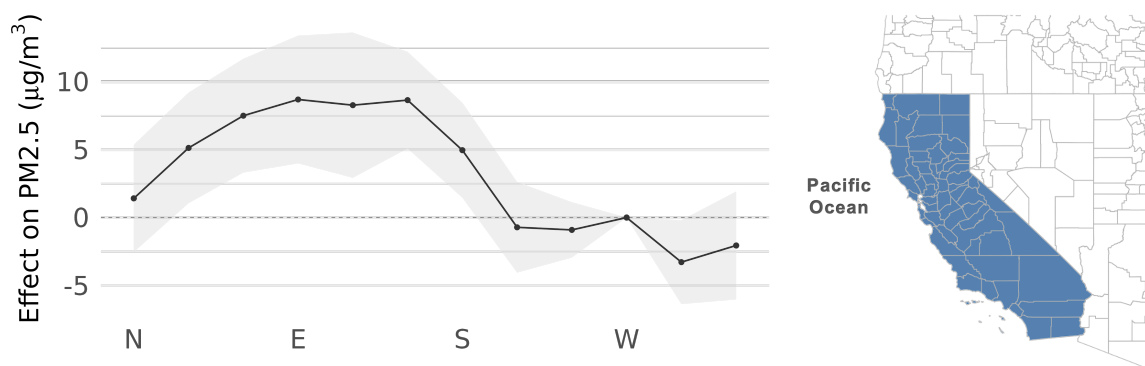


Figure 1. The relationship between wind direction and $\text{PM}_{2.5}$ concentration in California

Notes: The graph on the left plots the relationship between $\text{PM}_{2.5}$ concentration and wind direction in California. Wind direction describes where the wind is blowing *from*, with “N” indicating north, “E” indicating east, etc. The points report coefficient estimates from a regression of $\text{PM}_{2.5}$ on wind direction measured in 30-degree angle bins; wind from the west (“W”) is the omitted reference category. The shaded area shows the 95% confidence interval. All regressions include county-by-month, county-by-year, month-by-year, and day-of-week fixed effects, as well as flexible weather controls.

I interpret my IV estimate as a weighted average of treatment effects among compliers, with weights proportional to the magnitude of the first stage for each

the west, counties to the west of this source will record low pollution levels, and counties to the east will record high pollution levels. On net, a researcher who uses such variation may conclude that short-term pollution fluctuations have no effect on visit rates.

complier (Angrist et al., 2000). Figure A3 shows the geographic distribution of the strength of the first stage, measured as the difference in predicted PM_{2.5} levels between the most and least polluting wind directions.¹³

5 Empirical Results

5.1 Impact of Pollution on Economic Activity

I begin by estimating the effect of daily air pollution exposure on same-day activity, i.e., $k = 0$ in Equation (4). Table 2 displays estimates from both OLS and IV models. For the IV strategy, I use daily changes in county-level wind direction as an instrument for daily changes in county-level PM_{2.5} concentrations. The first-stage F-statistic in Column (2) is 87.36, which indicates that weak instrument concerns are minimal. Since PM_{2.5} is endogenous, I rely on the IV approach as the preferred empirical strategy. The estimate in Column (2) implies that a 1 $\mu\text{g}/\text{m}^3$ increase in PM_{2.5} reduces visit rates by approximately 0.20 visits per thousand people, equivalent to a 0.29% decrease relative to the mean daily visit rate. Columns (3)-(5) report the effects for the three largest sectors in the sample: retail trade, accommodation and food services, and entertainment and recreation. Column (5) shows that the relative effect is largest in the entertainment and recreation sector, nearly twice the overall average. This result is potentially because recreational activities are more flexible and easier to adjust or cancel.

The IV estimate in Column (2) is substantially larger than the OLS estimate in Column (1), suggesting that OLS estimation suffers from significant bias. This downward bias is common in quasi-experimental studies on air pollution and is generally thought to be, at least in part, due to measurement errors in pollution exposure (Schlenker and Walker, 2016; Ebenstein et al., 2017; Deryugina et al., 2019; Alexander and Schwandt, 2022).

¹³For each geographic cluster g , I calculate the predicted PM_{2.5} levels from $\hat{\pi}_b^g$ in Equation (5) for $b = 1, 2, 3, 4$ and take the difference between the maximum and minimum values.

Table 2. OLS and IV Estimates of Effect of PM2.5 on 1-day Visits

	OLS		IV		
	(1)	(2)	(3)	(4)	(5)
PM2.5 ($\mu g/m^3$)	-0.003*** (0.0014)	-0.20*** (0.01)	-0.06*** (0.005)	-0.04*** (0.005)	-0.03*** (0.003)
Relative Effect (%)	-0.004	-0.29	-0.27	-0.28	-0.47
Location Types	All	All	Retail Trade	Accommodation and Food Services	Entertainment and Recreation
First-stage F-stat		87.36	87.26	81.88	82.18
Dependent Variable Mean	69.85	69.85	22.40	14.49	6.41
Fixed Effects	Yes	Yes	Yes	Yes	Yes
R ²	0.87	0.86	0.90	0.85	0.73
Observations	4,501,200	4,495,000	4,487,988	4,472,910	4,304,146

Notes: This table reports the OLS and IV estimates using Equation (4) and (5). The dependent variable is number of visits per thousand people on the day of exposure. All regressions control for bins of mean temperature, precipitation, wind speed and dew point; two lags of these weather controls. IV estimates also include two lags and two leads of instruments. Fixed effects include county-by-year, county-by-month, day-of-week, and month-by-year FE. Column (1) and (2) show the effect for all POIs, whereas columns (3)-(5) are for three specific types of POIs. Standard errors are clustered at the county level. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Figure 2 extends this analysis by presenting percentage changes across all economic sectors.¹⁴ While most estimates are negative, the magnitudes vary, ranging from 0.47% in entertainment and recreation to 0.08% in health care. I find no statistically significant effect for several non-consumer-facing sectors, including manufacturing, waste management, public administration, professional and technical services, wholesale trade, and construction. This suggests that sectors less reliant on consumer foot traffic are less sensitive to air pollution.

My estimates for each sector are broadly consistent with the existing literature. Specifically, for recreation and entertainment, Fan (2024) finds that a 10 $\mu g/m^3$ increase in PM2.5 reduces outdoor activity by 1.4%, and Keiser et al. (2018) report a 3.9% decline in national park visits per 1-ppb increase in ozone; my estimates fall within this range. For food services and retail trade, Sun et al. (2019) find that a 10% increase in PM2.5 reduces restaurant visits by 0.15% and shopping visits by 0.1% in Beijing, which is similar in magnitude to my findings. For education, Currie et al. (2009) show that high CO increases absenteeism in elementary and middle schools; my estimate is in the same direction but smaller, possibly due to broader inclusion of

¹⁴Sectors are defined by 2-digit NAICS codes. The raw number of visits and each industry's share in the sample are reported in Table A1.

educational facilities. In healthcare, where prior evidence is mixed (see Table A2 for more details), I find a small decline in same-day visits but a statistically significant increase in monthly visits (Figure A8), suggesting short-term avoidance followed by delayed care.

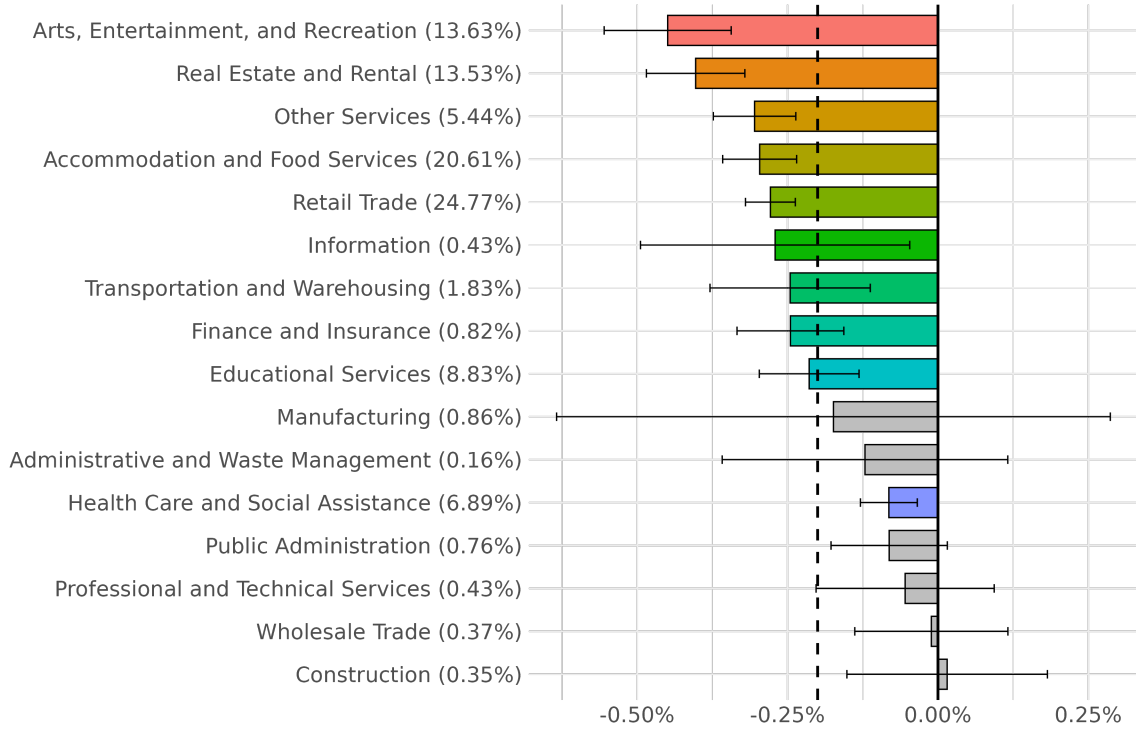


Figure 2. Relative effect (percent of 1-day visits), by 2-digit NAICS Codes

Notes: This figure displays the heterogeneous treatment effects of air pollution on daily activities across various industries, categorized by their 2-digit NAICS codes. Each bar represents an IV estimate from Equation (4) of the effect of 1-day PM2.5 exposure on same-day visits for a given economic sector. The percentage following each industry name indicates the share of raw visits that the industry represents in the sample. Bars represent the estimated effects, with horizontal lines indicating 95% confidence intervals. The vertical black dashed line shows the average effect from the main results across all POIs. Industries accounting for less than 0.2% of raw visits are omitted due to high variance. Sectors with statistically insignificant estimates are shown in gray. The corresponding absolute effects (in visits per thousand) are presented in Figure A4.

In addition, I provide further analysis of the most affected industry (i.e., entertainment and recreation) by examining more detailed facility types, defined using 6-digit NAICS codes.¹⁵ As shown in Figure A5a, air pollution significantly reduces visits to nature parks, golf courses, fitness centers, and museums, with estimated decreases ranging from 0.017 for nature parks to -0.0002 visits per thousand for museums.

¹⁵The share of each facility type in the sample is reported in Table A3.

When expressed in relative terms, however, these differences are much smaller and the reduction is all around 0.5% (Figure A5b). In contrast, visits to bowling centers slightly increase by about 0.27%.

I also estimate the effect of air pollution on the Healthcare and Social Assistance sector. As shown in Figure A6a, air pollution reduces visits to several types of healthcare facilities, such as nursing care, child day care, and specialty hospitals. This finding contrasts with the initial hypothesis that hospital visits would increase with worsening pollution. One possible explanation is the lack of a dedicated NAICS category for respiratory-related visits, which may obscure increases in emergency care within broader declines.¹⁶ Similarly, increases in visits by vulnerable groups may be offset by reductions in other healthcare services. To explore this possibility, I restrict the sample to counties with an elderly population above the median. As shown in Figure A6b, the estimates shift from negative to less negative or positive, with a notable increase in visits to general hospitals, the most common facility type in the sample.

My key identifying assumption is that changes in wind direction affect economic activity only through their impact on pollution levels. This assumption would be violated if wind direction were correlated with unobserved weather patterns that also influence daily activity. While this cannot be tested directly, I assess the plausibility of the assumption by examining how sensitive my estimates are to alternative ways of controlling for weather conditions, as shown in Table 8. Column (2) reports the estimate from a specification that omits all weather controls. Columns (3) and (4) report estimates using linear and quadratic weather controls, respectively, instead of the baseline weather bins. In Column (5), I coarsen the interior temperature bins to span 5 °C instead of 3 °C, and redefine dew point and wind speed bins from quintiles to quartiles. The estimated effect remains stable across all specifications, supporting the validity of the identifying assumption.

Figure 3 presents IV estimates of the effect of a one-day, 1 $\mu\text{g}/\text{m}^3$ increase in PM2.5 on economic activity over the month following exposure. The estimate at day 0 corresponds to the estimate from Column (2) of Table 2.¹⁷ If the cumulative effect recovers over time, this suggests that people are making up for lost activity rather than forgoing it entirely. In contrast, if the cumulative effect continues to decrease, it

¹⁶For example, Liu et al. (2022) find that air pollution increases the likelihood of refraining from healthcare visits in China.

¹⁷Estimates are converted to percentage terms using average daily visit rates to ensure comparability with previous results.

suggests that people may exhibit inertia and remain less active even after pollution levels improve.

I find a continuous decline in economic activity lasting up to two weeks, from approximately 0.29% on the first day to 1.27% after two weeks. Activity then gradually recovers, suggesting people partially make up for lost visits. By the end of one month, the cumulative effect is about two times larger than the contemporaneous effect, although estimates become less precise at longer horizons due to wider standard errors (Figure 3). This dynamic aligns with findings from [Barwick et al. \(2024\)](#), who document that air pollution significantly reduces spending on necessities and supermarket visits in China within two weeks, followed by a recovery thereafter.

This observed pattern is consistent with mechanisms documented in previous literature. Acute air pollution exposure can trigger adverse health effects and cause people to feel unwell ([Neidell et al., 2023](#); [Schlenker and Walker, 2016](#); [Deryugina et al., 2019](#)), potentially reducing their willingness or ability to engage in regular economic activities ([Graff Zivin and Neidell 2012](#); [Hanna and Oliva 2015](#)). Additionally, air pollution episodes have been shown to significantly increase household medical expenditures ([Barwick et al., 2024](#)), which could crowd out discretionary consumption. These combined health and economic impacts may explain the pronounced and persistent declines observed in discretionary sectors, such as recreation (Figure 4).

As a falsification test, Figure 3 also shows estimates of the effect of PM2.5 on cumulative activity in the two weeks *prior* to exposure. While there is some fluctuation, the pre-period estimates are small and show no clear trend, supporting the validity of my empirical strategy.

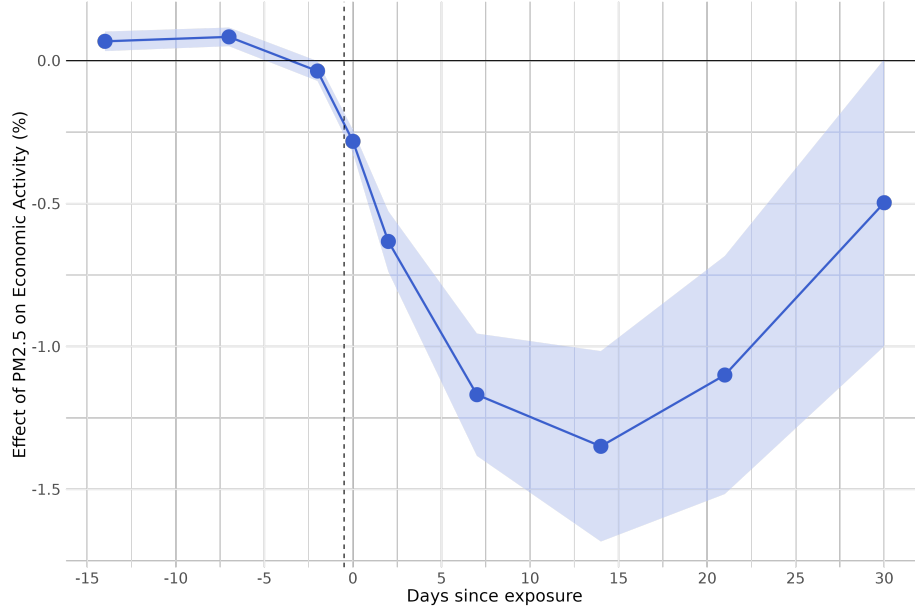


Figure 3. IV estimates of effect of acute (1-day) PM2.5 on cumulative economic activity

Notes: This figure displays IV estimates of the effects of a 1-day, $1 \mu\text{g}/\text{m}^3$ increase in PM2.5 on economic activities up to one month. Visit rates are measured as cumulative visits over a time window ranging from 1 to 30 days, as indicated on the x-axis. Points represent the point estimates, and the vertical lines represent the 95% confidence intervals. Standard errors are clustered by county.

Figure 4 shows how the estimated cumulative effects of a one-day PM2.5 increase vary across economic sectors, focusing on the three largest sectors in my sample: retail trade, accommodation and food services, and arts, entertainment, and recreation. Consistent with the contemporaneous results, the recreation sector (green line) experiences the largest reduction, both immediately and persistently. Over one month, the cumulative effects in the recreation sector deepen, becoming roughly nine times larger than the contemporaneous impacts. Retail trade (blue line) and accommodation and food services (orange line) exhibit similar magnitudes and trajectories. Although the cumulative effect in accommodation and food services slightly increases over one month, it stabilizes after the first week, reflecting a persistent rather than worsening impact. Overall, these results indicate lasting negative effects of acute pollution exposure, strongest in the recreation sector and more moderate in retail trade and accommodation services.

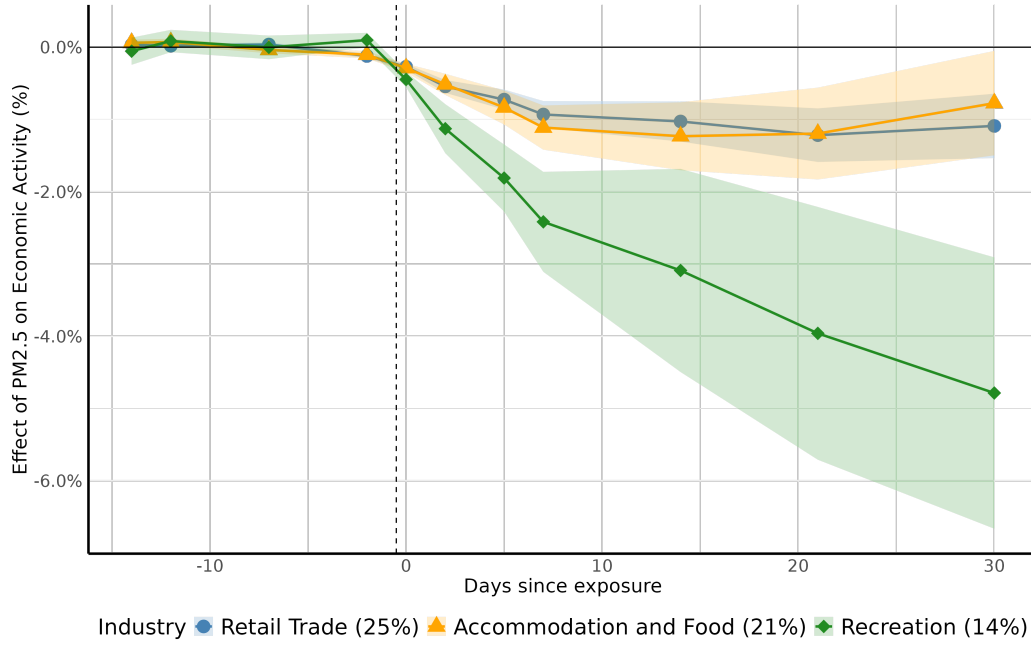


Figure 4. IV Estimates of the Effect of Acute (1-Day) PM2.5 Exposure on Cumulative Economic Activity Over One Month

Notes: This figure displays IV estimates of the effects of a 1-day, $1 \mu\text{g}/\text{m}^3$ increase in PM2.5 on economic activities up to one month. Visit rates are measured as cumulative visits over a time window ranging from 0 to 30 days, as indicated on the x-axis. Points represent the point estimates, and the vertical lines represent the 95% confidence intervals. Standard errors are clustered by county. Corresponding results in absolute terms (visits per 1,000 people) are shown in Figure A7.

Figure A8 presents sector-specific estimates of PM2.5 effects over time for additional industries. Sectors such as other services, finance and insurance, public administration, and information experience an immediate decline, but after one month, their recovery suggests that people eventually make up for lost activities in these sectors. In contrast, transportation and warehousing, wholesale trade, and retail trade exhibit both immediate and sustained declines, highlighting their vulnerability to air pollution.¹⁸

¹⁸Interestingly, I observe a contemporaneous decline followed by an increase in activities in real estate, educational services, and health care industries over one month. This may be due to: 1) In real estate, exposure to poor air quality may prompt people to seek housing in areas with better environmental conditions, increasing property visits over time (Chen et al., 2022; Pan, 2023). 2) Educational disruptions, such as pollution-induced school closures, can drive greater demand for alternative education services outside school (Currie et al., 2009). 3) In health care, the health effects of pollution exposure may take time to manifest (Simeonova et al., 2021).

5.2 Heterogeneity

A growing literature shows that exposure to air pollution and other environmental risks is unequally distributed across demographic groups (Mohai et al., 2009; Hsiang et al., 2019). To examine whether the effects of air pollution also differ across groups, I estimate the effects separately for counties grouped by income, age composition, and racial composition.

I begin with heterogeneity by income. Counties are divided into two groups based on whether their median income is above or below the national median.¹⁹ As shown in Figure 5a, high-income counties (darker solid line) experience larger declines in economic activity both contemporaneously and cumulatively over time. One possible explanation is that individuals in higher-income areas may be more aware of the health risks associated with pollution and are better able to adjust their behavior. Notably, since high-income counties have higher baseline visit rates (71.1 vs. 68.6 visits per 1,000 people), the absolute declines in visits are even larger.²⁰

I also examine heterogeneity by age composition, focusing on the share of children under five. Children are especially vulnerable to pollution due to ongoing lung development (Dietert et al., 2000; Aragón et al., 2017; Jayachandran, 2009). Similar to the income analysis, counties are grouped based on whether their under-five population share is above or below the median. As shown in Figure 5b, counties with more young children (darker solid line) experience larger and more persistent declines in economic activity, suggesting stronger avoidance behavior among communities with a higher proportion of vulnerable populations. Similarly, counties with a higher share of young children have higher baseline activity levels (76.6 vs. 63.1 visits per 1,000 people), implying a larger decline in visits in absolute terms.

Finally, I examine heterogeneity by racial composition. Counties are grouped based on whether their share of White residents is above or below the median. As shown in Figure 5c, counties with a higher White share (darker solid line) experience a sharper initial reduction in economic activity after pollution exposure. However, these counties also demonstrate a stronger recovery by the end of one month. Prior research indicates that air pollution disproportionately affects minority populations' health (Currie and Walker, 2011; Chay and Greenstone, 2003). One potential explanation is differential avoidance behavior: communities with a lower White share may engage

¹⁹See Appendix Table A5 for 1-day estimates, which include the interaction between income group and PM2.5 levels. Since PM2.5 is endogenous, the interaction term is instrumented using wind directions.

²⁰Results in absolute terms (visits per 1,000 people) are shown in Figure A9.

less frequently in pollution avoidance activities, resulting in prolonged exposure and potentially worse long-term health outcomes. It is worth noting that counties with a higher White share have lower baseline visit rates (63.1 vs. 76.6 visits per 1,000 people) but still experience slightly larger absolute reductions (Figure A9c).

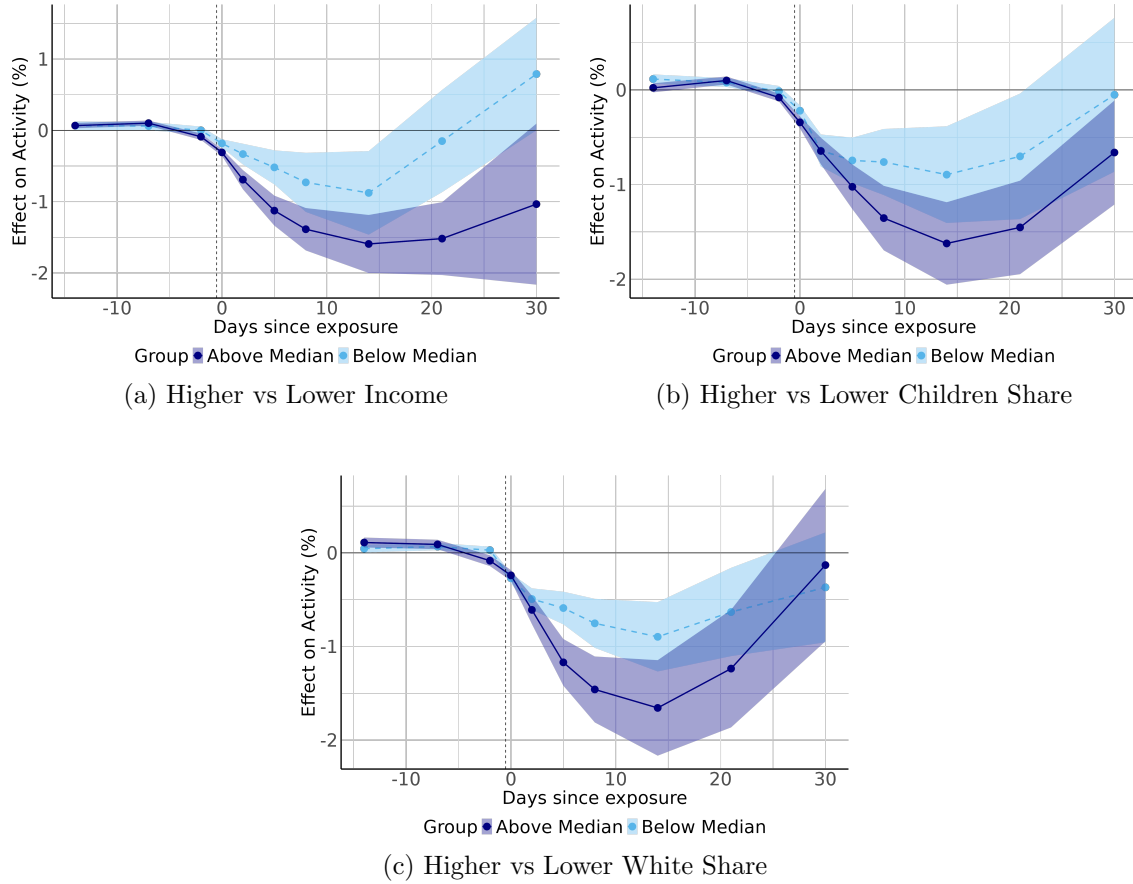


Figure 5. IV Estimates of Effect of Acute PM_{2.5} Exposure on Economic Activity, by demographic groups

Notes: Each point represents an IV estimate from Equation (4) for different subsamples, measuring the effect of acute PM_{2.5} exposure on economic activity in percentage terms across demographic groups. The solid line represents counties with income, children share, or White share above the median, while the dashed line represents those below the median. The cumulative effect is measured over one month, as indicated on the x-axis. Shaded areas denote 95% confidence intervals. All regressions include county-by-month, county-by-year, month-by-year, and day-of-week fixed effects, along with flexible controls for temperature, precipitation, dew point, and wind speed. Additionally, regressions incorporate leads of these weather controls, as well as two leads and two lags of the instruments. Standard errors are clustered at the county level. Results in absolute terms (visits per 1,000 people) are shown in Figure A7.

5.3 Mechanisms

In this section, I explore two potential mechanisms through which air pollution reduces daily activities. First, individuals may respond directly to pollution information, including national Air Quality Index (AQI) categories and Action Day alerts issued by local governments. A second potential mechanism is that individuals spend more time at home on days with poor visibility. Additionally, pollution-induced health impairments and increased medical expenditures are discussed based on insights from previous literature.

5.3.1 Pollution Information

AQI Categories AQI is a standardized measure ranging from 0 to 500, developed by the Environmental Protection Agency (EPA) to communicate air pollution levels to the public. Real-time AQI values and corresponding behavioral guidelines are disseminated through official channels, including the EPA’s website (www.airnow.gov) and dedicated mobile applications. Table A6 presents the AQI categories, their associated behavioral recommendations, and corresponding PM2.5 concentrations.

When PM2.5 levels exceed $35.5 \mu\text{g}/\text{m}^3$, corresponding to an AQI of 101, the information system color-codes the pollution level as orange, indicating that air quality is unhealthy for sensitive groups. Most weather apps and websites prominently display advisories under these conditions (see Figure A10). Although the marginal increase in pollution levels around this threshold is small, pollution information becomes much more salient to the public.

To estimate the causal effects of categorical pollution information and widely disseminated advisories on behavior, I utilize a regression discontinuity (RD) design with PM2.5 concentrations serving as the running variable. The analysis examines outcomes on either side of the $35.5 \mu\text{g}/\text{m}^3$ threshold. A crucial assumption for the validity of this approach is the absence of manipulation at this threshold, which is reasonable since PM2.5 data are automatically recorded by air quality monitors. Figure A11 supports this assumption. Given that AQI advisories are based on EPA monitor data, subsequent analyses employ monitor-based PM2.5 measurements.²¹

Table 3 summarizes RD estimates across various bandwidths and kernel specifications. The results indicate a negative but statistically insignificant effect of AQI advisories on visitation rates. Similarly, Figure 6 reveals no clear discontinuity at

²¹Due to the absence of monitors in some counties, the analysis includes 601,894 observations covering January 1, 2018, through December 30, 2021.

the threshold. This lack of significant findings could result from limited observations around the cutoff, as only 0.34% of data points surpass the AQI threshold of 100. Alternatively, individuals may respond only when advisories indicate much higher pollution levels. Previous research supports this notion, showing stronger behavioral responses at higher pollution thresholds (e.g., AQI of 300).²²

Table 3. Estimation Results from RD of AQI Advisories

	(1)	(2)	(3)	(4)	(5)	(6)
AQI Advisories	-4.0 (3.9)	-3.0 (2.7)	-2.1 (1.9)	-3.0 (3.8)	-3.2 (2.6)	-1.4 (1.8)
Kernel	Triangular	Triangular	Triangular	Uniform	Uniform	Uniform
Bandwidth	5	10	20	5	10	20
Dependent Variable Mean	75.99	75.99	75.99	75.99	75.99	75.99
Observations	691,387	691,387	691,387	691,387	691,387	691,387
Effective Observations	1,939	5,744	42,595	1,987	5,863	43,567

Notes: This table presents regression discontinuity (RD) estimates examining the effects of AQI advisories on economic activity, using PM2.5 as the running variable with a cutoff at $35.5 \mu g/m^3$ (equivalent to AQI = 101, unhealthy for sensitive groups). The dependent variable is visits per thousand people on the day of exposure. Estimates are shown for different bandwidth choices and kernel specifications.

²²For example, [Neidell \(2009\)](#) and [Zivin and Neidell \(2009\)](#) demonstrate significant responses to smog alerts issued for ozone levels equivalent to AQI 300 in California.

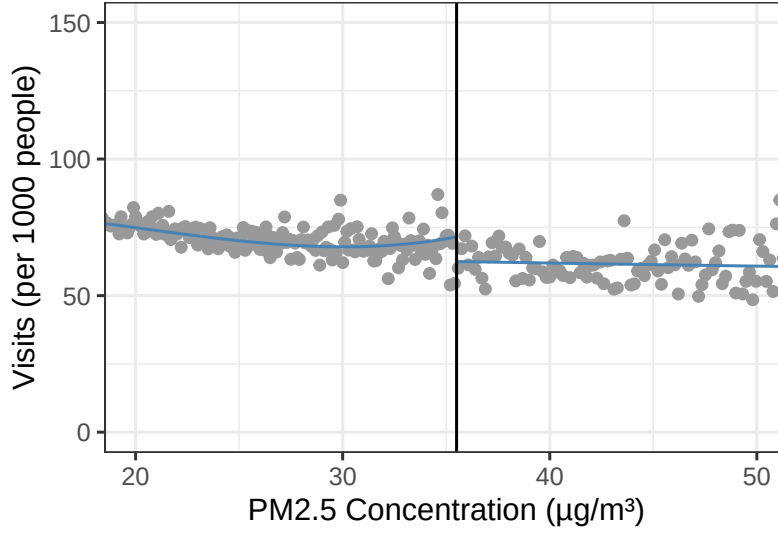


Figure 6. Visits per 1,000 People by PM2.5 Levels

Notes: This figure presents a regression discontinuity plot showing the relationship between daily PM2.5 concentrations and visits per 1,000 people. The vertical line at $35.5 \mu\text{g}/\text{m}^3$ represents the EPA-defined threshold for air quality being unhealthy for sensitive groups (AQI = 101). Points represent binned averages of visitation rates, and solid lines are polynomial fits estimated separately on each side of the threshold.

Action Day Alerts Unlike uniform AQI advisories, Action Day alerts are discretionary warnings issued by local air quality management agencies when air pollution is forecasted to reach unhealthy levels. On Action Days, governments urge the public to reduce pollution by minimizing driving, limiting outdoor activities, and staying indoors. The criteria and timing for Action Day alerts differ substantially across jurisdictions, providing natural variation for evaluating their effectiveness in influencing behavioral responses.

To systematically analyze the effects of Action Day alerts, I obtain a comprehensive dataset from the EPA’s AirNow program, covering 346 reporting jurisdictions across cities, counties, metropolitan areas, and states, representing approximately 50 percent of the US population. To ensure consistency and avoid jurisdictional duplication, data are aggregated at the Core-Based Statistical Area (CBSA) level.

Figure A12 demonstrates that the probability of Action Day alerts increases with rising daily PM2.5 concentrations. However, fewer than one-fifth of high-pollution days trigger Action Day alerts, providing variation to examine whether these alerts independently influence economic activity, conditional on similar pollution levels.

Table 4 presents empirical results examining the incremental effect of Action Day

alerts on economic activity. Although the estimated magnitude of Action Day alerts is approximately twice that of the PM2.5 effect alone, the estimates are imprecise and statistically insignificant, likely due to the small share (around 1.8%) of Action Days. As an additional robustness check, I also estimate effects excluding observations with consecutive Action Days, and the results remain very similar (Columns (3) and (4) of Table 4).

Table 4. Additional Effects of Action Days on Economic Activity

	All Days		Exclude Consecutive Action Days	
	(1)	(2)	(3)	(4)
PM2.5 ($\mu g/m^3$)	-0.10*** (0.02)	-0.09*** (0.02)	-0.12*** (0.03)	-0.12*** (0.03)
Action Day Alerts		-0.22 (0.32)		-0.15 (0.26)
First-stage F statistic	12.2	12.2	14.7	14.7
Dependent variable mean	79.8	79.8	80.0	80.0
Fixed effects	Yes	Yes	Yes	Yes
R ²	0.88	0.88	0.87	0.87
Observations	428,352	428,352	422,983	422,983

Notes: This table reports IV estimates from Equations (4) and (5), examining the additional effects of Action Day advisories. The dependent variable is visits per thousand people at the CBSA level on the day of exposure. All regressions include CBSA-by-month, CBSA-by-year, day-of-week, and month-by-year fixed effects, as well as flexible weather controls. Column (1) shows the effect of instrumented PM2.5 only; Column (2) adds controls for Action Day alerts. Columns (3) and (4) repeat the analysis excluding consecutive Action Day sequences. Standard errors clustered at the CBSA level are reported in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

5.3.2 Visibility

The same pollutants that contribute to PM2.5 can also reduce visibility: airborne particles impair visibility by altering how light is absorbed and scattered in the atmosphere, reducing the clarity and color of what we see.²³ When visibility is low, people might perceive that the air quality is poor and choose to spend more time at home instead of going out.

To examine whether reduced visibility serves as a mechanism through which pollution affects economic activity, I use satellite-based visibility data from the NCEP

²³Source: https://www.epa.gov/sites/default/files/2015-05/documents/haze_brochure_20060426.pdf. Accessed June 10, 2024.

North American Regional Reanalysis (NARR) database.²⁴ I implement a three-step procedure. First, I instrument PM2.5 using wind direction, as in Equation (5). Second, I test whether PM2.5 causally reduces visibility by estimating:

$$\text{Visibility}_{ct} = \rho \cdot \widehat{\text{PM2.5}}_{ct} + \mathbf{X}'_{ct}\gamma + \sigma_{cy} + \eta_{cm} + \mu_w + \psi_{my} + \epsilon_{ct} \quad (6)$$

Finally, I regress visit rates on visibility while controlling for instrumented PM2.5. This allows me to examine whether visibility, after controlling for the direct effect of PM2.5, provides additional explanatory power for economic activity. However, since visibility itself is not instrumented, this approach provides correlational evidence rather than causal estimates regarding the relationship between reduced visibility and lower economic activity.²⁵ Specifically, I estimate:

$$\frac{Y_{ct}}{\text{Pop}_c} = \lambda \cdot \text{Visibility}_{ct} + \kappa \cdot \widehat{\text{PM2.5}}_{ct} + \mathbf{X}'_{ct}\gamma + \sigma_{cy} + \eta_{cm} + \mu_w + \psi_{my} + \nu_{ct} \quad (7)$$

The results are presented in Table 5. Column (1) shows that PM2.5 significantly reduces visibility: a $1 \mu\text{g}/\text{m}^3$ increase in PM2.5 leads to a 0.2 km decrease in visibility on average.²⁶ Column (2) indicates that reduced visibility is correlated with lower daily economic activity: a 1 km reduction in visibility is associated with a decrease of 0.17 visits per 1,000 people on average. These findings are consistent with Keiser et al. (2018), who find that visitation to national parks declines on days with poor visibility.

²⁴See <https://psl.noaa.gov/data/gridded/data.narr.html>. Accessed July 18, 2024.

²⁵Controlling for instrumented PM2.5 helps account for other behavioral or physiological channels through which pollution may directly affect economic activity.

²⁶To account for the possibility of a nonlinear relationship between PM2.5 and visibility, I repeat the analysis using the logarithm of visibility as the outcome. Results, presented in Table A7, remain consistent in direction and magnitude.

Table 5. Visibility as a Mechanism: Effect of PM2.5 on Visibility and Visits

	(1)	(2)
	Visibility (<i>km</i>)	Visits (per 1,000 people)
Visibility (<i>km</i>)		0.168*** (0.005)
$\widehat{PM2.5}$ ($\mu g/m^3$)	-0.016** (0.007)	-0.124*** (0.014)
First-stage F stat	87.36	87.36
Dependent Variable Mean	17.55	69.85
Fixed Effects	Yes	Yes
R ²	0.53	0.87
Observations	4,501,200	4,501,200

Notes: This table reports results from the three-step mechanism analysis. Column (1) shows the effect of instrumented PM2.5 on visibility (Equation (6)), and Column (2) shows the effect of visibility on visit rates, controlling for instrumented PM2.5 (Equation (7)). All regressions include county-by-month, county-by-year, day-of-week, and month- by-year fixed effects, as well as flexible weather controls. The dependent variable mean in Column (1) is average visibility (in kilometers), and in Column (2) is average visits per 1,000 people. Standard errors are clustered at the county level.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

5.3.3 Health and Income Effects

Beyond responses to air quality alerts and visible environmental cues, air pollution may also reduce daily activity through direct health effects and household budget constraints. While empirically testing these mechanisms is beyond the scope of this paper, existing literature offers substantial insights.

Exposure to elevated PM2.5 levels can cause physical discomfort and symptoms such as fatigue, coughing, or chest tightness. A growing body of evidence finds that pollution worsens short-term health outcomes, including respiratory and cardiovascular complications (Neidell et al., 2023; Schlenker and Walker, 2016; Deryugina et al., 2019; He et al., 2020; Moretti and Neidell, 2011). These acute health shocks may prompt individuals to avoid public spaces or cancel non-essential activities, especially those that require physical effort or take place in crowded environments. I find that a 1 $\mu g/m^3$ increase in PM2.5 reduces same-day visit rates by 0.29%, a magnitude comparable to health-related responses documented in other settings. For example, He et al. (2020) find that a 1 $\mu g/m^3$ increase in PM2.5 from straw burning raises mortality by 0.33%, and Barwick et al. (2024) find that a 1 $\mu g/m^3$ increase in PM2.5 leads to a 0.07% increase in healthcare transactions in the short run and 0.27% over

the medium term. These patterns suggest that even moderate pollution episodes may impose a health burden or discomfort that plausibly contribute to the activity reductions I estimate.

Pollution may also indirectly reduce activity by tightening household budget constraints. If pollution increases medical needs, the resulting rise in out-of-pocket spending may reduce discretionary consumption. As noted above, Barwick et al. (2024) estimate that a $1 \mu\text{g}/\text{m}^3$ increase in PM2.5 leads to a 0.27% increase in healthcare transactions over the medium term. Given that healthcare comprises roughly 8% of household budgets, this translates to approximately a 0.02% increase in total consumption.²⁷ In contrast, I estimate a 0.5% decline in visit rates over one month. This gap suggests that the financial burden from increased medical costs likely plays a minor role in explaining the overall reduction in activity.

The combination of temporary illness and increased medical costs offers a plausible explanation for the sustained decline in visitation following pollution exposure, particularly in discretionary sectors like recreation (see Figure 4). Health-related incapacitation may limit individuals' ability to go out even after air quality improves, while elevated medical spending may reduce available resources for non-essential consumption. Beyond these two channels, other mechanisms may also contribute. For example, temporary avoidance during polluted periods may give rise to habit formation or behavioral inertia, leading individuals to maintain lower activity levels even after pollution returns to normal.

5.4 Welfare Analysis

In this study, I present novel evidence on individuals' daily adjustments in response to air pollution. A $1 \mu\text{g}/\text{m}^3$ increase in PM2.5 leads to a 0.29% reduction in economic activities across all industries on average. The behavioral responses I observe are relatively smaller compared to existing studies on avoidance behaviors. For example, Zivin and Neidell (2009) find that ozone alerts reduce visits to zoos and observatories in Los Angeles by 5–8%, and He et al. (2016) find that an API above 100 reduces movie theater admissions by 2.26%.

This relatively smaller magnitude is expected for two reasons. First, prior studies often focus on discretionary activities like zoo visits or movie attendance, which are

²⁷According to the Consumer Expenditure Survey (2022), healthcare spending constitutes approximately 8% of total household consumption. See https://www.bls.gov/news.release/archives/cesan_09082023.pdf, accessed August 17, 2025.

more easily postponed or canceled. Second, prior studies often evaluate responses to air quality alerts, which prompt stronger behavioral reactions. My findings, capturing adjustments to everyday pollution, are most comparable to Fan (2024), who show that a $10 \mu g/m^3$ increase in PM2.5 reduces outdoor exercise by 1.4%.

This study reveals an important social cost of air pollution that has been overlooked in previous research. Air pollution imposes costs not only through direct health consequences but also through economic losses from daily behavioral changes. In this section, I put the results into context by generating some back-of-the-envelope estimates of the economic costs associated with the decline in daily activities due to air pollution. Specifically, I estimate an OLS regression of county-level per capita income on daily activity levels, controlling for county and year fixed effects:

$$\log(Income_{cy}) = \alpha \cdot \log\left(\frac{Y_{cy}}{Pop_c}\right) + \tau_c + \phi_y + \epsilon_{cy} \quad (8)$$

where $Income_{cy}$ is the personal income per capita in county c in year y , and $\frac{Y_{cy}}{Pop_c}$ is the average visit rates in county c in year y . The regression also controls for county fixed effects (τ_c) and year fixed effects (ϕ_y). The coefficient of interest, α , represents the association between county daily activities and county income per capita. The estimated coefficient is 0.018 with a standard error of 0.004, indicating that a 1% decrease in daily activities is associated with a 0.018% decrease in personal income per capita.

In the previous section, I show that a $1 \mu g/m^3$ increase in PM2.5 decreases daily activities by 0.29% on average in the short term (1-day window), translating into a 0.0052% reduction in personal income per year. Given the average personal income per capita of \$64,665 in the US, this percentage reduction corresponds to an income loss of \$3.36 per person annually, or approximately \$1.1 billion nationwide.²⁸ Extending the analysis over a one-month horizon, the cumulative decline in daily activities is larger at 0.47%, implying an annual personal income loss of approximately 0.0085%, equivalent to \$5.50 per person or roughly \$1.8 billion nationwide.²⁹ Since daily activities also contribute to physical health (e.g., reduced obesity) and psychological well-being (e.g., reduced depression), these estimates likely represent a lower bound of the total welfare impacts.

To address potential endogeneity between daily activity and income, I also conduct an IV analysis using annual rainfall as an instrument. The significant positive IV

²⁸The World Bank reported a total US population of 331.9 million in 2021.

²⁹Note that this estimate is less precise (significant at the 10% level) due to wider standard errors.

estimate (0.3) reinforces the conclusion that higher daily activity is associated with higher income. Although the estimate is substantially larger than the OLS estimate (0.018), its interpretation depends on the assumption that rainfall affects income only through daily activity. Since it is difficult to fully rule out other channels, such as effects on agriculture or infrastructure, I interpret the IV result with caution and continue to rely on the OLS estimate for welfare calculations.

The estimated welfare impact of air pollution on daily activities is relatively minor compared to other air pollution cost estimates. For example, the World Bank estimated that the welfare cost of air pollution in the United States was around \$886.5 billion in 2016 (\$1,080.96 billion in 2022 USD). However, these two estimates are not directly comparable. The World Bank’s estimate encompasses comprehensive costs related to human health and environmental damage, whereas my estimate specifically quantifies economic costs arising only from daily behavioral adjustments. As people become more educated about the impacts of pollution, the costs associated with the reduction in daily activities are expected to increase. However, the cost of pollution related to mortality and morbidity might decrease due to more widespread adoption of avoidance behaviors.

A more meaningful comparison would be with two recent studies on avoidance behaviors. [Zhang and Mu \(2018\)](#) estimates the cost of defensive expenditures on face masks during heavily polluted periods to be \$187 million USD, and [Fan \(2024\)](#) estimates the cost of physical inactivity, as it relates to major non-communicable diseases, to be \$0.55 billion USD in 2017. My cost estimates are larger despite smaller declines in daily activity, for two main reasons: First, previous studies focus on the costs of specific avoidance behaviors, such as purchasing face masks or reducing outdoor exercise, whereas my estimate captures a more general cost. Second, previous analyses examine the costs associated with heavily polluted days, which typically occur only a few times a year, while my analysis considers behavioral adjustments to everyday pollution, which occurs consistently throughout the year.

6 Robustness Checks

In this section, I first test the validity of the IV. IV estimates can be interpreted as the local average treatment effect (LATE) when the monotonicity assumption holds ([Angrist and Imbens, 1995](#)). In this paper, this assumption will be satisfied if every county within a geographic cluster group experiences a change in pollution in the same

direction when the wind blows from a 90-degree direction bin, and will be violated if some counties experience changes in different directions with other counties within the same cluster group. One way [Deryugina et al. \(2019\)](#) assess the validity of this assumption is by varying the number of geographic clusters and the sizes of the wind direction bins. I follow a similar approach by changing the number of geographical clusters from 20 to 10 and 30 and reducing the size of wind angle bins from 90 degrees to 60 degrees. In Table 6, Column 1 is the original specification, Column 2 and 3 change the number of geographical clusters, and Column 4, 5, and 6 decrease the size of wind angle bins. In all cases, the IV estimates are similar to the main specification, supporting the robustness of the main result to different instrument choices. In addition, I address potential spatial autocorrelation by clustering the standard errors at the county-day level. As shown in Table A8, the estimates remain significant.

Table 6. Robustness of IV Estimates to Instrument Choices

	(1)	(2)	(3)	(4)	(5)	(6)
PM2.5 ($\mu g/m^3$)	-0.20*** (0.01)	-0.22*** (0.01)	-0.18*** (0.01)	-0.25*** (0.01)	-0.23*** (0.01)	-0.17*** (0.01)
Number of geographic clusters	20	10	30	10	20	30
Size of wind angle bins	90°	90°	90°	60°	60°	60°
R ²	0.86252	0.86123	0.86	0.86	0.86	0.86
Observations	4,495,000	4,495,000	4,495,000	4,495,000	4,495,000	4,495,000
F-statistic	87.36	143.36	60.04	93.36	57.51	49.48
Dependent Variable Mean	69.85	69.85	69.85	69.85	69.85	69.85

Notes: This table reports the IV estimates using Equation (4) and Equation (5) when varying the instrument choices. The dependent variable is visits per thousand people on the day of exposure. All regressions include county-by-month, county-by-year, day-of-week, and month-by-year fixed effects, as well as flexible weather controls. The baseline model (shown in column (1)) aggregates location into 20 clusters and wind direction into 90-degree intervals. Standard errors clustered at the county level are reported in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Another underlying assumption of this IV approach is that the variation comes primarily from the pollution that is transported by wind rather than generated locally. If this underlying assumption holds, then the first stage should be generally weak on days with low wind speeds and vice versa. To further examine the validity of this IV approach, I calculate the first-stage F-statistics separately by quintiles of daily wind speed. As shown in Figure 7, the strength of the first stage increases as wind speed increases. This implies the pollution variation is mainly due to non-local transport by wind, which supports the validity of my approach.

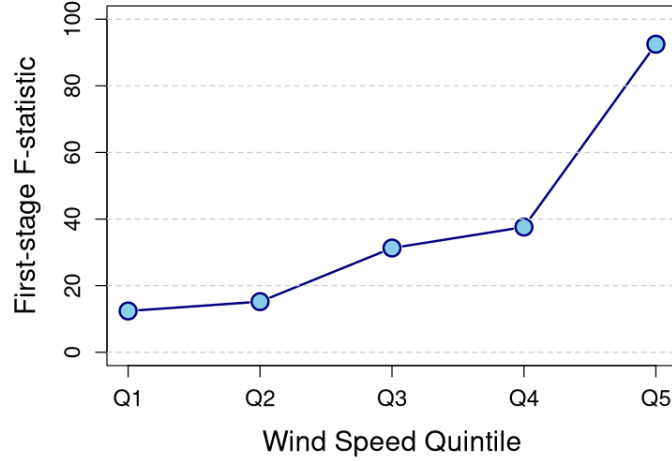


Figure 7. Relationship Between the First-stage F Statistics and Wind Speed

Notes: This figure displays the First-stage F Statistics for five subsamples that each include days that fall within a particular wind speed quintile. The first-stage F-statistics are generally smaller on days with low wind speeds and bigger on days with high wind speeds.

In addition, Table 7 indicates that the main specification is robust to variations in the number of instrument lags included. This demonstrates that the main estimates are not driven by lagged effects from PM2.5 on previous days, and therefore, can be properly interpreted as the impact of a one-unit increase in daily PM2.5 levels.

Table 7. Robustness of IV Estimates to Including Different Instrument Lags

	(1)	(2)	(3)	(4)	(5)
	1 lead and 1 lag	1 lag	2 lags	3 lags	4 lags
PM2.5 ($\mu\text{g}/\text{m}^3$)	-0.20*** (0.01)	-0.13*** (0.01)	-0.20*** (0.01)	-0.20*** (0.01)	-0.20*** (0.01)
# of instruments leads and lags	2	0	1	3	4
First-stage F-stat	87.36	162.90	87.36	77.71	84.56
R ²	0.86	0.86	0.8	0.86	0.86
Observations	4,495,000	4,501,200	4,495,000	4,488,800	4,482,600
Dependent Variable Mean	69.85	69.85	69.85	69.85	69.85

Notes: This table reports the IV estimates using Equation (4) and Equation (5) when including different number of instrument lags. The baseline model (shown in column (1)) include 2 leads and 2 lags. The dependent variable is visits per thousand people on the day of exposure. All regressions include county-by-month, county-by-year, day-of-week, and month-by-year fixed effects, as well as flexible weather controls. Standard errors clustered at the county level are reported in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Then, I check the robustness of the main specification along several dimensions. First, Table 9 displays estimates using alternative sets of fixed effects, confirming that

the main results are robust and not driven by seasonal or regional patterns. Second, Table 8 shows results from alternative weather control specifications, implying that the estimates are not driven by climatic factors correlated with wind direction and economic activity. Lastly, Table A9 presents estimates using the log and inverse hyperbolic sine transformations instead of raw visit rates, implying that the findings are insensitive to alternative functional forms of the dependent variable.

Table 8. Robustness of IV Estimates to Including Different Forms of Weather Controls

	(1)	(2)	(3)	(4)	(5)
PM2.5 ($\mu g/m^3$)	-0.20*** (0.01)	-0.14*** (0.007)	-0.12*** (0.01)	-0.17*** (0.01)	-0.16*** (0.01)
First-stage F-stat	87.36	262.53	87.52	80.82	95.85
Dependent Variable Mean	69.85	69.85	69.85	69.85	69.85
Fixed Effects	Yes	Yes	Yes	Yes	Yes
R ²	0.86	0.86	0.87	0.86	0.86
Observations	4,495,000	4,495,000	4,495,000	4,495,000	4,495,000
Weather Controls					
Baseline	✓				
Linear			✓		
Quadratic				✓	
Less Granular bins					✓

Notes: This table reports IV estimates from Equation (4) and Equation (5) using different combinations of weather controls. The dependent variable is the number of visits per thousand people on the day of exposure. All regressions include county-by-year, county-by-month, day-of-week, and year-by-month fixed effects. Column (1) presents my baseline specification, controlling for bins of mean temperature, precipitation, wind speed, dew point, and all observed combinations of these variables. Column (2) omits all weather controls. Column (3) uses linear weather controls instead of bins. Column (4) uses quadratic weather controls. Column (5) uses 5-degree Celsius temperature bins instead of 3-degree bins. Standard errors clustered at the county level are reported in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Table 9. Robustness of IV Estimates to Including Different Forms of Fixed Effects

	(1)	(2)	(3)	(4)	(5)	(6)
PM2.5 ($\mu\text{g}/\text{m}^3$)	-0.20*** (0.01)	-0.23*** (0.01)	-0.16*** (0.05)	-0.16** (0.06)	-0.21*** (0.06)	-0.27*** (0.08)
County-by-year FE	✓	✓	✓			
County-by-month FE	✓	✓				
Month-by-year FE	✓		✓	✓		✓
Day-of-week FE	✓		✓	✓		
State-by-month FE				✓	✓	
State-by-year FE				✓	✓	✓
R ²	0.86252	0.79508	0.80735	0.31468	0.24577	0.26824
Observations	4,495,000	4,495,000	4,495,000	4,495,000	4,495,000	4,495,000
F-statistic	87.363	84.767	50.032	62.051	60.244	43.394
Dependent Variable Mean	69.85	69.85	69.85	69.85	69.85	

Notes: This table reports IV estimates from Equation (4) and Equation (5) using different combinations of fixed effects. The dependent variable is the number of visits per thousand people on the day of exposure. Column (1) presents my baseline specification. Standard errors clustered at the county level are reported in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

I have thus far interpreted my estimates as the causal effects of exposure to PM2.5. One potential concern, however, is that other harmful pollutants may be co-transported with PM2.5, potentially confounding the estimated effect. To address this issue, I re-estimate the main specification while including daily concentrations of SO₂, NO₂, and O₃ as additional controls. Table 10 reports the results. The coefficient on PM2.5 remains negative, and statistically significant, indicating that the main findings are not driven by co-movements with other pollutants.

Table 10. Robustness of IV Estimates to Controlling for Other Pollutants

	(1)	(2)	(3)	(4)	(5)
PM2.5 ($\mu g/m^3$)	-0.20*** (0.01)	-0.18*** (0.01)	-0.15*** (0.02)	-0.13*** (0.02)	-0.13*** (0.02)
SO ₂ (ppb)		0.009 (0.06)		-0.08 (0.06)	-0.03 (0.08)
NO ₂ (ppb)			-0.07** (0.03)		-0.03 (0.04)
O ₃ (ppb)				0.12*** (0.01)	0.12*** (0.01)
R ²	0.86	0.86	0.86	0.86	0.87
Observations	4,495,000	4,495,000	4,495,000	4,495,000	4,495,000
F-statistic	87.36	87.36	87.36	87.36	87.36
Dependent Variable Mean	69.85	69.85	69.85	69.85	69.85

Notes: This table reports IV estimates from Equation (4) and Equation (5) when controlling for additional air pollutants. The dependent variable is the number of visits per thousand people on the day of exposure. Column (1) presents the baseline specification without additional pollutants. Standard errors clustered at the county level are reported in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

For a final robustness check, I estimate the regression using different subsamples. The sample period ranges from January 1, 2018 to December 31, 2021, which includes the COVID-19 pandemic, which dramatically affected individuals' mobility patterns. To ensure that the results are not influenced by public-health guidance on economic activity, I estimate the effect separately before and after the outbreak of the COVID-19 pandemic.³⁰ As shown in Table A10, the estimates are very similar, indicating the negative impact is not driven by pandemic-related restrictions. In addition, Table A11 shows the robustness of the main specification when excluding counties without satellite data.

7 Conclusion

This paper provides the first large-scale analysis of how PM2.5 affects daily economic activities across the United States. Leveraging changes in wind direction as an instrumental variable, I address the endogeneity of air pollution exposure and find robust evidence that PM2.5 significantly reduces daily visitation rates. Specif-

³⁰The cutoff date for dividing the sample is mid-March 2020, when many states and cities in the U.S. began implementing lockdowns, stay-at-home orders, and other restrictions to curb the spread of the virus.

ically, a $1 \mu\text{g}/\text{m}^3$ increase in PM_{2.5} decreases visit rates by an average of 0.29%, translating into a nationwide annual reduction of approximately 24.6 million trips and an economic cost exceeding \$1.1 billion. The effects are widespread across sectors, particularly pronounced in recreation and entertainment, and vary significantly across demographic groups. Wealthier counties and those with larger shares of young children exhibit stronger behavioral responses, suggesting differential abilities or willingness to engage in avoidance behavior.

This study has several limitations. First, the analysis primarily captures short- and medium-term behavioral responses, leaving long-term adaptations unexplored. Second, while aggregated at the county level, the data obscures important individual-level heterogeneity in responses driven by factors such as health status, age, or socioeconomic conditions. Third, the reduced-form empirical strategy captures a composite effect of air pollution on economic activity, combining both active avoidance decisions and involuntary reductions driven by immediate health impacts, making it difficult to isolate these separate channels.

Despite these limitations, this study makes several important contributions. It provides comprehensive, nationally representative estimates of air pollution’s causal impact on economic activity, significantly extending previous findings typically limited to specific locations or activity types. Additionally, the findings underscore the critical importance of considering behavioral avoidance responses in pollution damage estimates, highlighting that ignoring these behaviors likely understates true social costs. Finally, this research emphasizes the broader welfare implications of pollution-driven reductions in daily activities, including losses in physical and psychological well-being and disruptions to key economic sectors. Policymakers should thus consider not only the direct health impacts of pollution but also the substantial economic and social costs arising from individuals’ daily adjustments to poor air quality.

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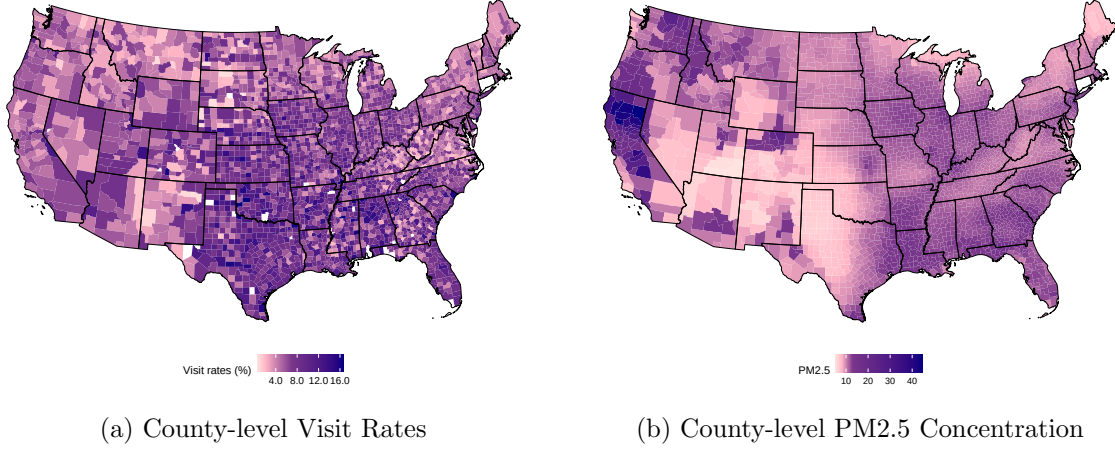
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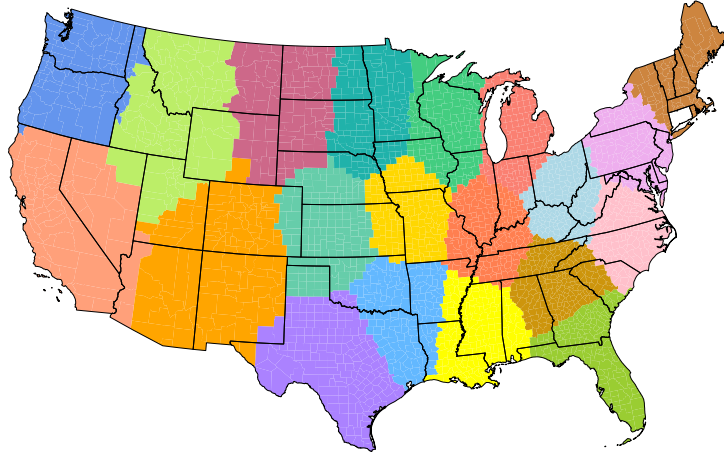
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A Appendix and Figures Tables



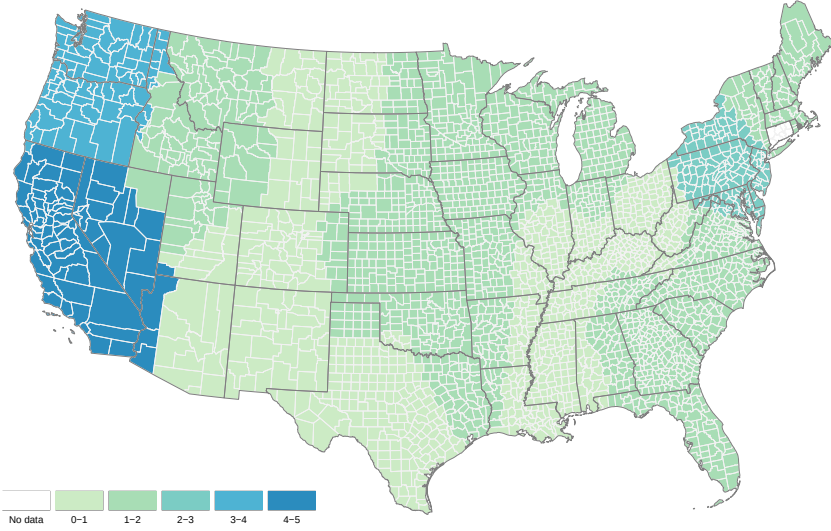
Appendix Figure A1. County-level visit rates and PM2.5 concentration

Notes: This figure displays average daily county average for the number of visits (left panel) and PM2.5 concentration (right panel) from January 1, 2018, to December 30, 2021. As a few counties do not have any visitation data, there are some missing values in the figure.



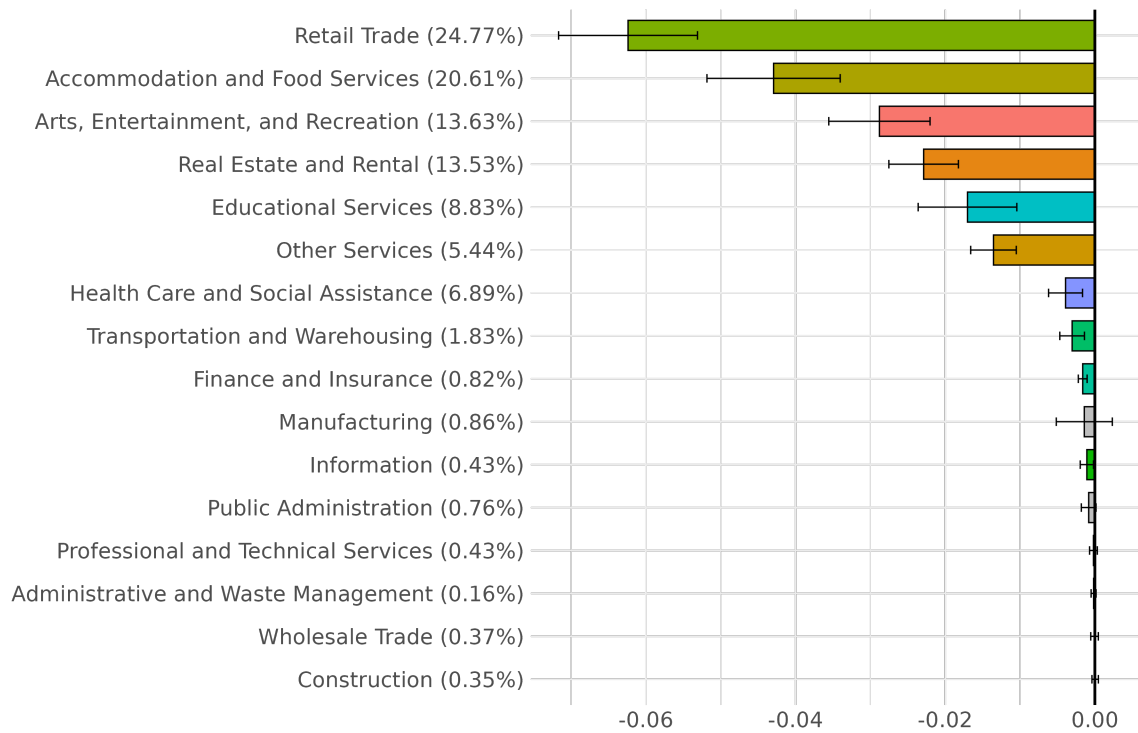
Appendix Figure A2. K-means Clustering Result

Notes: This figure displays the K-means clustering result based on latitude and longitude. As a few counties do not have any visitation data for leisure facilities from SafeGraph, there are some missing values in the figure. There are 20 spatial groups in total, and each of them is represented by a different color. After clustering, π_b^g in Equation (5) can vary across geographic regions.



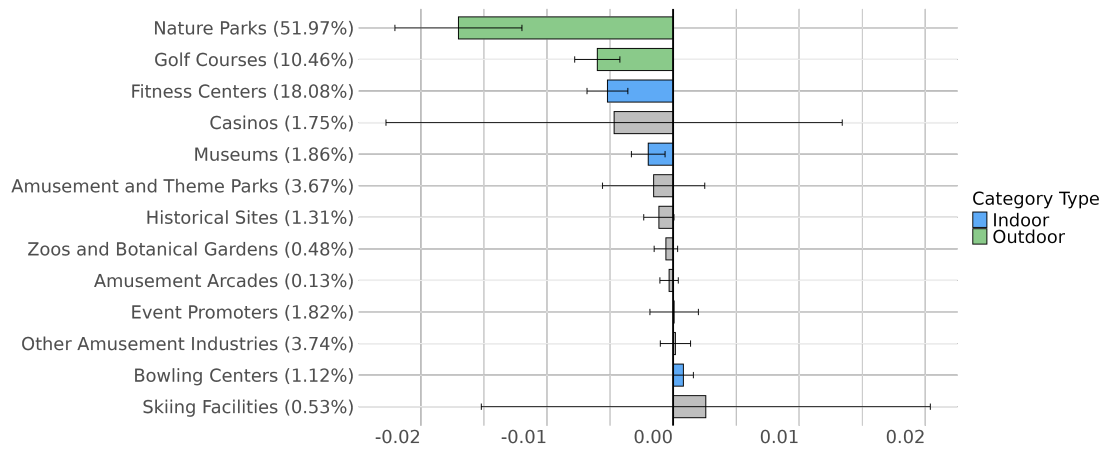
Appendix Figure A3. Strength of the First Stage, by Geographic Group

Notes: This map shows the strength of the first stage for each of the 20 geographic groups used in the main estimation sample. The strength is measured as the difference in predicted $\text{PM}_{2.5}$ concentrations ($\mu\text{g}/\text{m}^3$) between the most and least polluting wind direction bins, where predictions are obtained from the first-stage specification in Equation (5). The 20 geographic clusters are plotted in Figure A2.

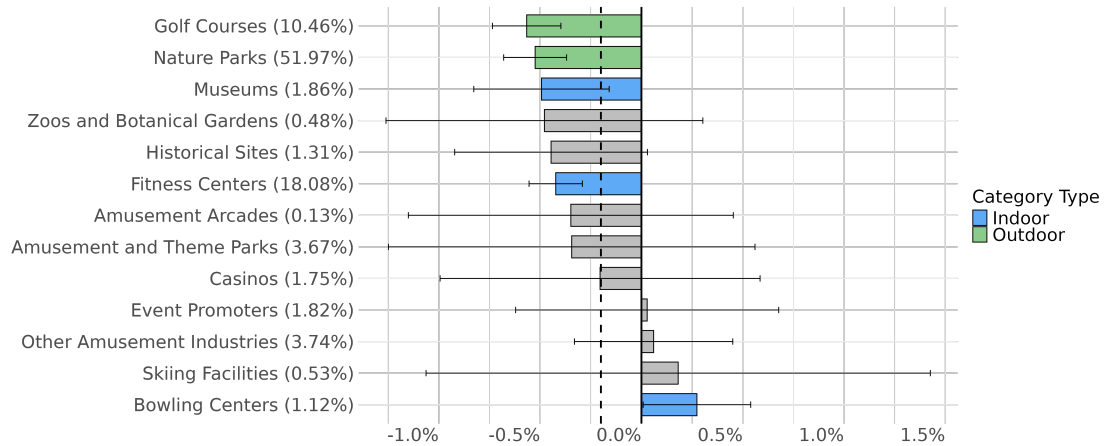


Appendix Figure A4. Absolute Effect (Visits per Thousand) by 2-Digit NAICS Code

Notes: This figure displays the heterogeneous treatment effects of air pollution on daily activities across various industries, categorized by their 2-digit NAICS codes. Each bar represents an IV estimate from Equation (4) of the effect of 1-day PM_{2.5} exposure on 1-day visits for a particular economic sector. The percentage following each industry name indicates the share of raw visits that the industry represents in the sample. Bars represent the estimated effects, with horizontal lines indicating 95% confidence intervals. Industries with raw visits accounting for less than 0.2% are omitted from the figure due to large variance. Sectors with statistically insignificant estimates are shown in gray. The relative effect as a percentage of the sector's average 1-day visits are shown in Figure 2.



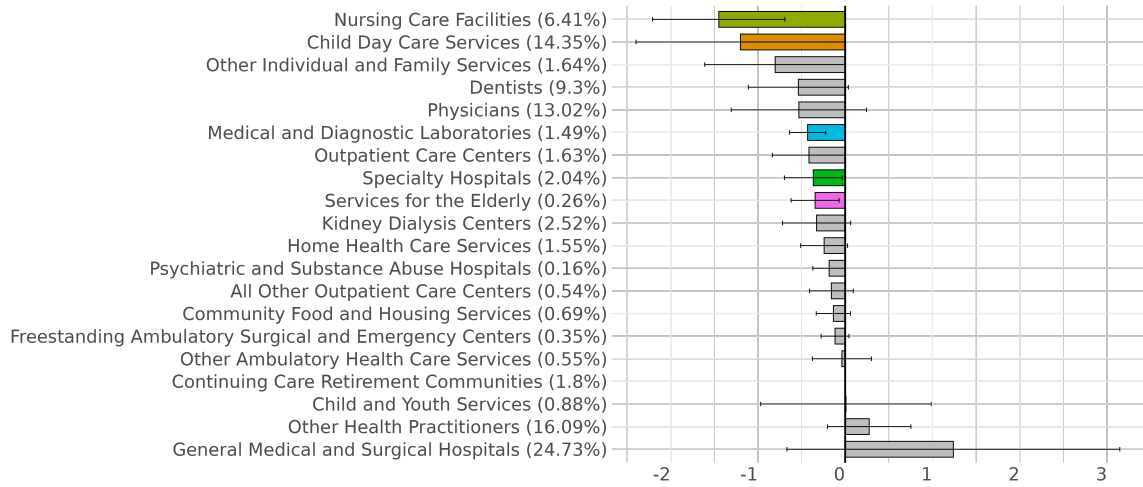
(a) Absolute Effect (Visits per Thousand)



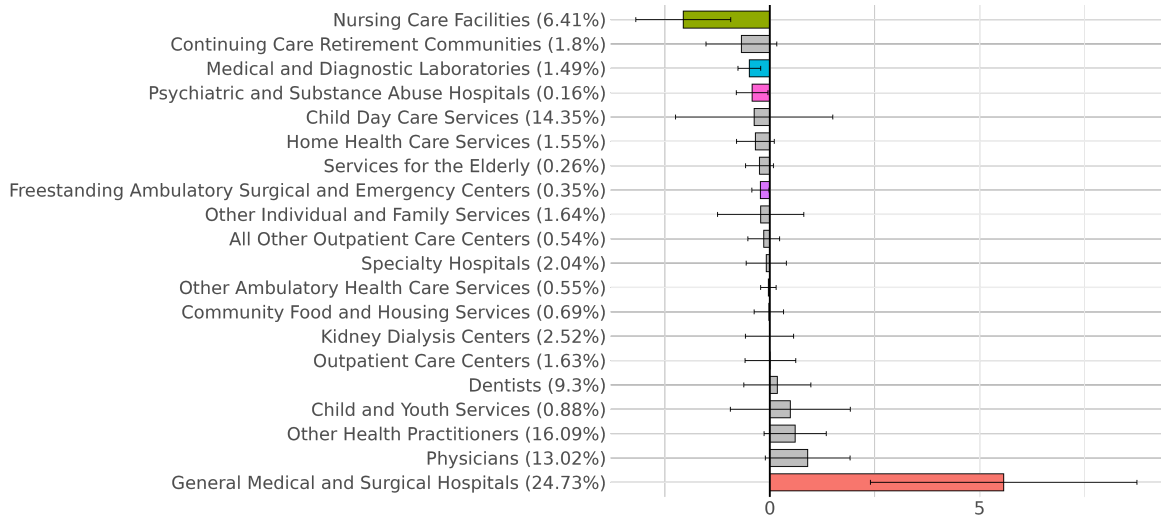
(b) Relative Effect (Percent of 1-day Visits)

Appendix Figure A5. Heterogeneous Effects of Air Pollution on Visits to Recreational Facilities

Notes: This figure displays the heterogeneous treatment effect of air pollution on visit rates at different recreational facilities, defined by 6-digit NAICS codes. Panel (a) shows the absolute effect in visits per thousand, and Panel (b) presents the relative effect as a percentage of same-day visits. Outdoor facilities (in green) include golf courses and nature parks, while indoor facilities (in blue) include museums, fitness centers, and bowling centers. Categories with statistically insignificant effects are shown in gray. Each bar represents an estimated effect, with horizontal lines indicating 95% confidence intervals. The vertical blue dashed line in panel (b) represents the average effect for all POIs. Categories accounting for less than 0.1% of raw visits are omitted due to high variance.



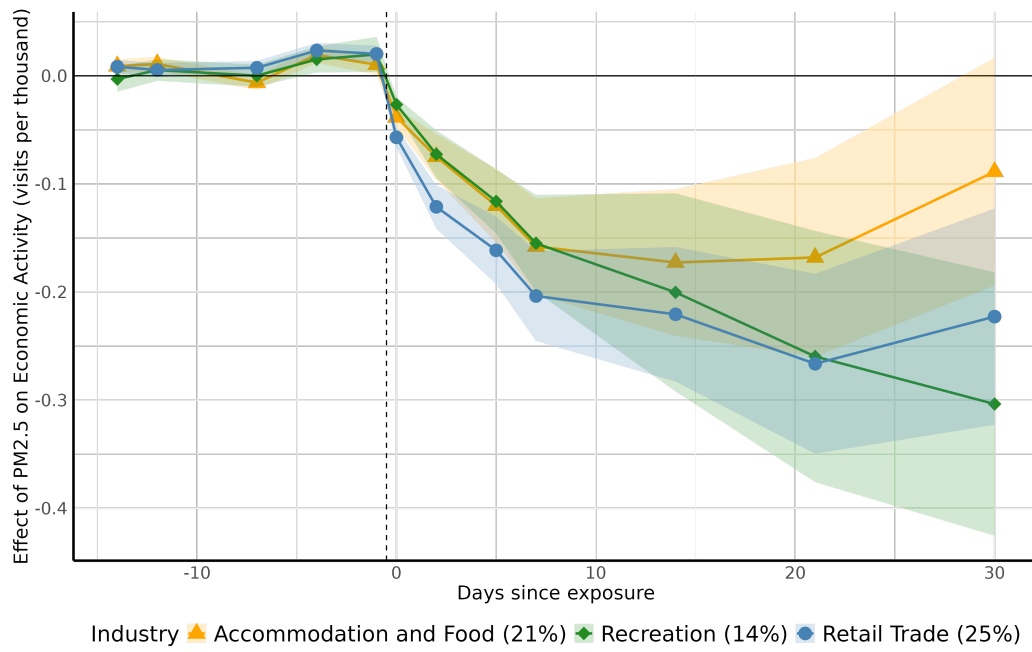
(a) Absolute Effect for All Counties



(b) Absolute Effect for Counties with Elderly Population Share Above the Median

Appendix Figure A6. Heterogeneous Effects of Air Pollution on Visits to Healthcare Facilities

Notes: This figure displays the heterogeneous treatment effect of air pollution on visit rates to different healthcare facilities, defined by 6-digit NAICS codes. Panel (a) presents the absolute effect (visits per thousand) for all counties, while Panel (b) shows the effect for counties with an elderly population share above the median. Each bar represents an estimated effect, with horizontal lines indicating 95% confidence intervals. Categories with statistically insignificant effects are shown in gray. Categories accounting for less than 0.1% of raw visits are omitted due to high variance.



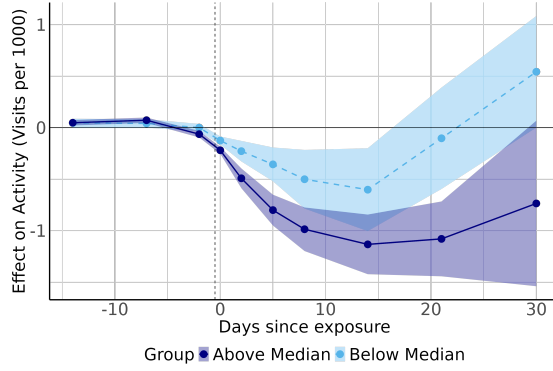
Appendix Figure A7. IV Estimates of the Effect of Acute (1-Day) PM2.5 Exposure on Cumulative Visits per 1,000 People Over One Month

Notes: This figure displays IV estimates of the effects of a 1-day, $1 \mu\text{g}/\text{m}^3$ increase in PM2.5 on economic activities, measured as cumulative visits per 1,000 people over a time window ranging from 0 to 30 days, as indicated on the x-axis. Points represent the point estimates, and vertical lines represent 95% confidence intervals. Standard errors are clustered by county. Corresponding results in percentage terms are presented in Figure 4.

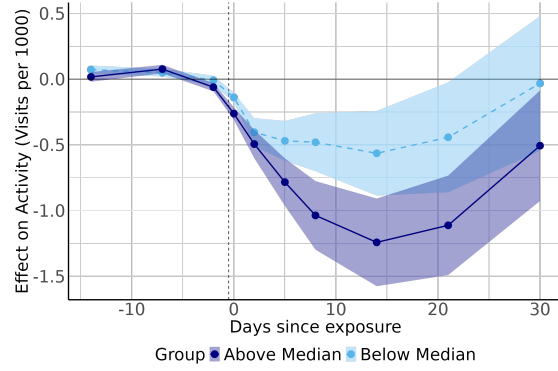


Appendix Figure A8. IV Estimates of effect of acute PM2.5 exposure on economic activity, by industries

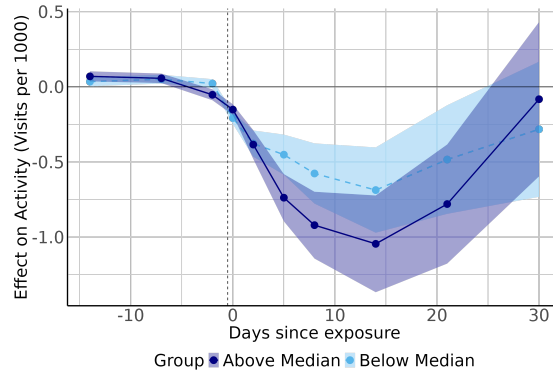
Notes: Each point represents an IV estimate from Equation (4) and (5), measuring the effect of acute PM2.5 exposure on economic activity in percentage terms across 15 different industries. The cumulative effect is measured over a time window ranging from 1 to 28 days, as indicated by the x-axis. Shaded areas represent 95% confidence intervals. All regressions include county-by-month, county-by-year, month-by-year, and day-of-week fixed effects, as well as flexible controls for temperature, precipitation, and wind speed. Additionally, the regressions incorporate leads of these weather controls, as well as one lead and one lag of the instruments. Standard errors are clustered by county.



(a) Higher vs Lower Income



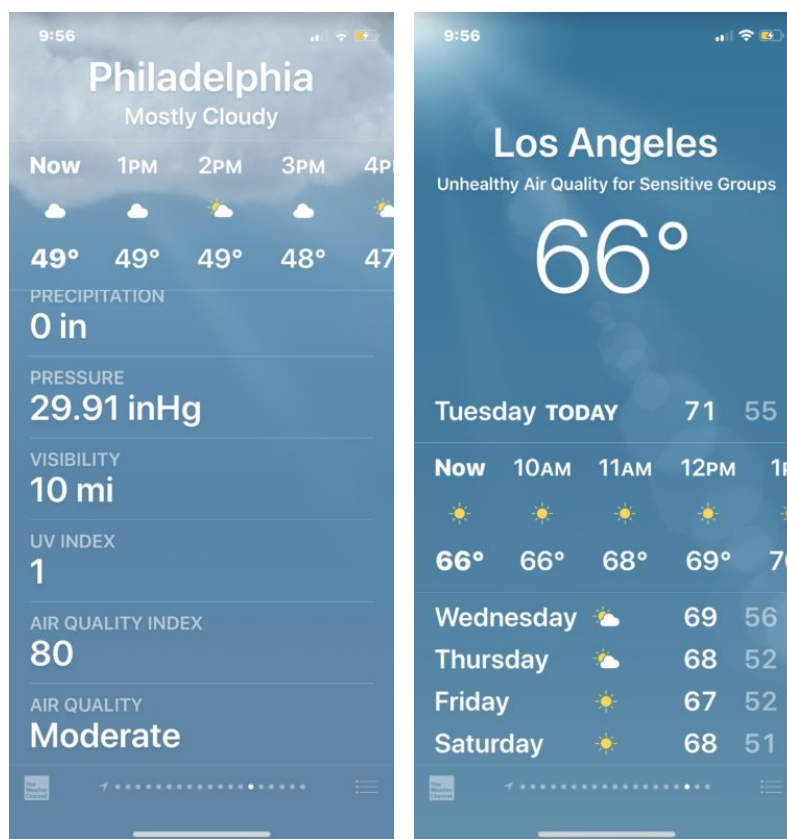
(b) Higher vs Lower Children Share



(c) Higher vs Lower White Share

Appendix Figure A9. IV Estimates of Acute PM2.5 Exposure on Economic Activity by Demographic Group (Absolute Changes)

Notes: Each point represents an IV estimate from Equation (4) for different subsamples, measuring the effect of acute (1-day) PM2.5 exposure on economic activity in absolute terms (visits per 1,000 people) across demographic groups. Solid line represents counties with median income, child population share, or White population share above the national median; the dashed lines represent counties below the median. The cumulative effect is tracked over one month following exposure, as indicated on the x-axis. Shaded areas denote 95% confidence intervals. All regressions include county-by-month, county-by-year, month-by-year, and day-of-week fixed effects, along with flexible controls for temperature, precipitation, dew point, and wind speed. Additionally, regressions incorporate leads of these weather controls, as well as two leads and two lags of the instruments. Standard errors are clustered at the county level.



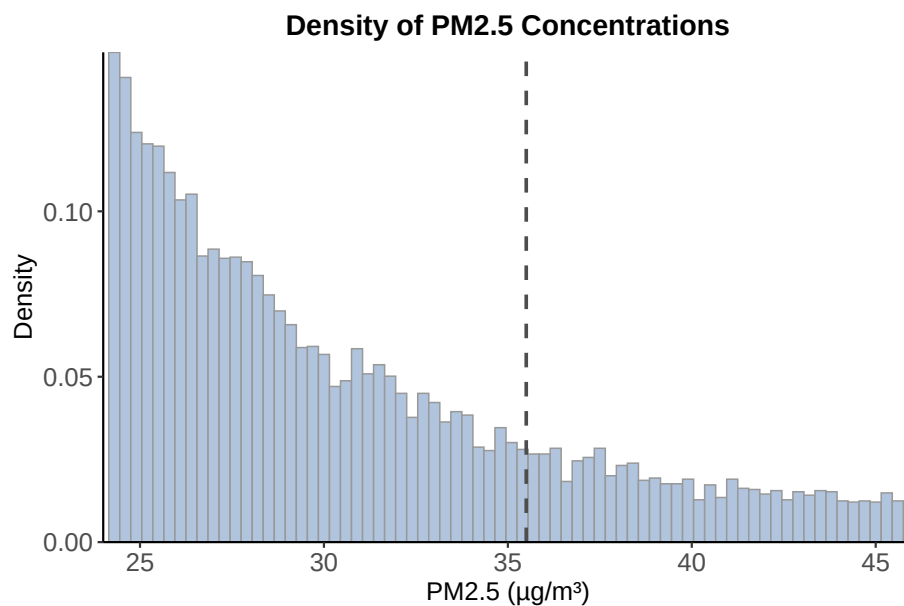
(a) when $AQI \leq 100$

(b) when $AQI > 100$

Appendix Figure A10. Example: Air Quality Information on iPhone with Weather

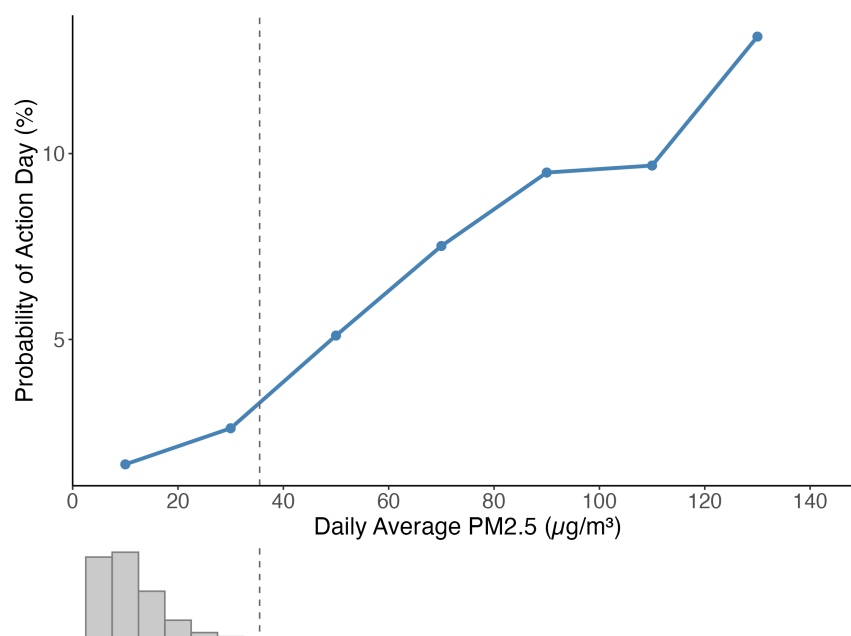
Notes: This figure displays the weather application interface when $AQI \leq 100$ (left panel) and $AQI > 100$ (right panel). Specifically, if there is no air quality concern ($AQI \leq 100$), then there is no message at the top of the Weather overview. Users have to scroll down to find the air quality information. However, if the air quality is unhealthy ($AQI > 100$), then the app prominently displays information regarding unhealthy air quality at the top of the interface.

Source: <https://osxdaily.com/2018/11/20/get-air-quality-info-iphone-weather/>



Appendix Figure A11. No Manipulation at the Threshold

Notes: This figure displays the density of daily PM_{2.5} concentrations. The vertical dashed red line at 35.5 µg/m³ indicates the EPA-defined threshold for air quality being “unhealthy for sensitive groups” (AQI = 101). The smooth distribution around this cutoff suggests no evidence of manipulation at the threshold.



Appendix Figure A12. Probability of Action Day Issuance Around PM_{2.5} Threshold

Notes: This figure plots the share of county-days with an Air Quality “Action Day” advisory by daily average PM_{2.5}. The vertical dashed line at 35.5 $\mu\text{g}/\text{m}^3$ indicates the EPA-defined threshold for air quality being “unhealthy for sensitive groups”. The lower histogram shows the distribution of PM_{2.5} observations in the sample.

Appendix Table A1. Percentage of Raw Visits by 2-digit NAICS Codes

NAICS Codes	Description	Raw Visit	Percentage
44-45	Retail Trade	9,146,929,016	24.77
72	Accommodation and Food Services	7,610,742,956	20.61
71	Arts, Entertainment, and Recreation	5,034,071,398	13.63
53	Real Estate and Rental and Leasing	4,996,089,773	13.53
61	Educational Services	3,261,648,196	8.83
62	Health Care and Social Assistance	2,543,964,413	6.89
81	Other Services	2,010,628,936	5.44
48-49	Transportation and Warehousing	676,689,537	1.83
31-33	Manufacturing	317,021,279	0.86
52	Finance and Insurance	303,659,378	0.82
92	Public Administration	279,409,983	0.76
51	Information	160,409,689	0.43
54	Professional, Scientific, and Technical Services	158,697,298	0.43
42	Wholesale Trade	136,210,405	0.37
23	Construction	130,696,514	0.35
55	Management of Companies and Enterprises	72,815,213	0.20
56	Administrative and Support and Waste Management	59,912,819	0.16
22	Utilities	28,890,736	0.08
11	Agriculture, Forestry, Fishing and Hunting	2,093,013	0.01
21	Mining, Quarrying, and Oil and Gas Extraction	185,256	0.00

Appendix Table A2. Literature review on the impact of air pollution on activities

Literature	Outcome	Explanatory Variable	Result
Entertainment & Recreation			
Fan (2024)	Outdoor exercise in China	10 $\mu g/m^3$ increase in PM2.5	1.4% reduction
Keiser et al. (2018)	Visits to U.S. national parks	1-ppb increase in ozone	3.9% reduction
Zivin and Neidell (2009)	Visits to zoos and observatory in LA	Ozone alert	5–8% reduction
Janke (2014)	Visits to Bristol Zoo (members)	Air pollution alert	6% reduction
Retail Trade			
Sun et al. (2019)	Visits to shopping centers in Beijing	10% increase in PM2.5	0.1% reduction
Addoum et al. (2023)	Visits to retail establishments in the US	Medium-heavy smoke day	0.37% reduction
Accommodation & Food Services			
Sun et al. (2019)	Visits to restaurants in Beijing	10% increase in PM2.5	0.15% reduction
Gao et al. (2020)	Visits to restaurants in Beijing	1% increase in PM2.5	0.065% reduction
Information			
He et al. (2016)	Movie theater admissions in China	Air pollution index	2.8% reduction
Educational Services			
Currie et al. (2009)	School absenteeism in Texas	Additional high-CO day	5–9% increase
Chen et al. (2000)	School absenteeism in Washoe County	50-ppb increase in ozone	13% increase
Healthcare			
Liu et al. (2022)	Visits to health facilities in China	1 $\mu g/m^3$ increase in PM2.5	1.8% reduction
Dardati et al. (2024)	ER visits in Chile	1 $\mu g/m^3$ increase in PM2.5	0.03–0.07% increase
Janke (2014)	Hospital admissions in the UK	1% increase in ozone	0.1% increase
Ren et al. (2021)	Hospital admissions in Wuhan, China	10% increase in PM2.5	1.25% increase

Appendix Table A3. Percentage of Raw Visits in Recreation Industry

NAICS Codes	Description	Raw Visit	Percentage
712190	Nature Parks and Other Similar Institutions	2,610,937,377	51.97
713940	Fitness and Recreational Sports Centers	908,175,772	18.08
713910	Golf Courses and Country Clubs	525,323,866	10.46
713990	All Other Amusement and Recreation Industries	187,873,033	3.74
713110	Amusement and Theme Parks	184,452,834	3.67
711211	Sports Teams and Clubs	126,947,155	2.53
712110	Museums	93,538,708	1.86
711310	Promoters of Performing Arts and Events	91,428,501	1.82
713210	Casinos (except Casino Hotels)	87,878,401	1.75
712120	Historical Sites	65,689,209	1.31
713950	Bowling Centers	56,054,357	1.12
713920	Skiing Facilities	26,600,467	0.53
712130	Zoos and Botanical Gardens	24,056,950	0.48
713930	Marinas	17,077,159	0.34
713120	Amusement Arcades	6,400,510	0.13
713290	Other Gambling Industries	4,858,269	0.10
711212	Racetracks	4,849,757	0.10
711219	Other Spectator Sports	915,133	0.02
711130	Musical Groups and Artists	528,696	0.01
711190	Other Performing Arts Companies	364,952	0.01
711510	Independent Artists, Writers, and Performers	70,944	0.00
711110	Theater Companies and Dinner Theaters	33,414	0.00
711410	Agents and Managers for Artists	30,905	0.00

Appendix Table A4. Percentage of Raw Visits in Healthcare Industry

NAICS Codes	Description	Raw Visit	Percentage
6221	General Medical and Surgical Hospitals	602,393,987	24.73
6213	Offices of Other Health Practitioners	391,980,708	16.09
6244	Child Day Care Services	349,496,920	14.35
6211	Offices of Physicians	317,171,391	13.02
6212	Offices of Dentists	226,457,069	9.30
6231	Nursing Care Facilities (Skilled Nursing Facilities)	156,062,572	6.41
621492	Kidney Dialysis Centers	61,429,935	2.52
6223	Specialty (except Psychiatric and Substance Abuse) Hospitals	49,666,480	2.04
6233	Assisted Living Facilities for the Elderly	43,721,328	1.80
624190	Other Individual and Family Services	39,871,875	1.64
6214	Outpatient Care Centers	39,677,848	1.63
6216	Home Health Care Services	37,830,065	1.55
6215	Medical and Diagnostic Laboratories	36,241,509	1.49
624110	Child and Youth Services	21,330,836	0.88
6242	Community Food and Housing, and Emergency	16,862,249	0.69
6219	Other Ambulatory Health Care Services	13,277,127	0.55
621498	All Other Outpatient Care Centers	13,132,241	0.54
621493	Freestanding Ambulatory Surgical and Emergency Centers	8,481,763	0.35
624120	Services for the Elderly and Persons with Disabilities	6,366,018	0.26
6222	Psychiatric and Substance Abuse Hospitals	3,974,312	0.16
6241	Individual and Family Services	161,226	0.01
621420	Outpatient Mental Health and Substance Abuse Centers	139,374	0.01

Appendix Table A5. Heterogeneous Effect of PM2.5 across Different Groups

	(1)	(2)
Panel A: Different Income Group		
PM2.5 ($\mu g/m^3$)	-0.08*** (0.03)	-0.11*** (0.02)
PM2.5 $\times$ 1{Income > Median}	-0.18*** (0.03)	
PM2.5 $\times$ 1{Income > 3rd Quartile}		-0.25*** (0.04)
Panel B: Different Age Group		
PM2.5 ($\mu g/m^3$)	-0.10*** (0.02)	-0.10** (0.04)
PM2.5 $\times$ 1{Pct. Children > Median}	-0.24*** (0.05)	
PM2.5 $\times$ 1{Pct. Children > 3rd Quartile}		-0.36*** (0.08)
Panel C: Different Race Group		
PM2.5 ($\mu g/m^3$)	-0.16*** (0.02)	-0.16*** (0.02)
PM2.5 $\times$ 1{Pct. White > Median}	-0.09*** (0.03)	
PM2.5 $\times$ 1{Pct. White > 3rd Quartile}		-0.16*** (0.04)
First-stage F stat	83.533	83.533
Dependent Variable Mean	69.85	69.85
Fixed Effects	Yes	Yes
R ²	0.87	0.87
Observations	4,493,550	4,493,550

Notes: This table reports the effect of daily PM2.5 on daily activities for different income groups using Equation (4) and Equation (5). The dependent variable is number of visits per thousand people on the day of exposure. All regressions control for bins of mean temperature, precipitation, wind speed and dew point; two lags of these weather controls. Fixed effects include county-by-year, county-by-month, day-of-week, and year-by-month FE. The dummy variable $\mathbf{1}\{\text{Income} > \text{Median}\} = 1$ ($\mathbf{1}\{\text{Income} > 3\text{rd Quartile}\} = 1$) if personal income per capita in county c is higher than the national median (third Quartile). Similarly, $\mathbf{1}\{\text{Pct. Children} > \text{Median}\}$ and $\mathbf{1}\{\text{Pct. Children} > 3\text{rd Quartile}\}$ are dummies for the percentage of children, and $\mathbf{1}\{\text{Pct. White} > \text{Median}\}$ and $\mathbf{1}\{\text{Pct. White} > 3\text{rd Quartile}\}$ are dummies for the percentage of the White population. Standard errors are clustered at the county level. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Appendix Table A6. AQI Categories corresponding to PM2.5 Concentrations

Category	Designated Color	AQI Index	PM2.5 Concentration ($\mu g/m^3$)
Good	Green	0-50	0.0-12.0
Moderate	Yellow	51-100	12.1-35.4
Unhealthy for Sensitive Groups	Orange	101-150	35.5-55.4
Unhealthy	Red	151-200	55.5-150.4
Very Unhealthy	Purple	201-300	150.5-250.4
Hazardous	Maroon	301-500	250.5-500

Source: National Ambient Air Quality Standards for Particle Pollution Fact Sheet. Available at: https://www.epa.gov/sites/default/files/2016-04/documents/2012_aqi_factsheet.pdf. Accessed February 25, 2023.

Appendix Table A7. Visibility as a Mechanism: Effect of PM2.5 on Log Visibility and Visits

	(1) log(Visibility)	(2) Visits (per 1,000 people)
log(Visibility)		0.9053*** (0.0475)
$\widehat{PM2.5}$ ($\mu g/m^3$)	-0.0027*** (0.0009)	-0.1257*** (0.0143)
First-stage F stat	163.8	163.8
Dependent Variable Mean	17.52	69.85
Fixed Effects	Yes	Yes
R ²	0.40	0.87
Observations	4,501,200	4,501,200

Notes: This table reports results from the three-step mechanism analysis. Column (1) shows the effect of instrumented PM2.5 on the logarithm of visibility (Equation (6)). Column (2) shows the effect of log(visibility) on visit rates, controlling for instrumented PM2.5 (Equation (7)). All regressions control for temperature, precipitation, wind speed, and dew point, including two lags of each weather variable. The dependent variable mean in Column (1) refers to visibility in kilometers (prior to log transformation), and in Column (2) is the average number of visits per 1,000 people. Fixed effects include county-by-year, county-by-month, day-of-week, and month-by-year. Standard errors are clustered at the county level. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Appendix Table A8. Robustness to Different Clustering Levels

	(1)	(2)	(3)	(4)
PM2.5 ($\mu g/m^3$)	-0.20*** (0.01)	-0.20*** (0.07)	-0.20*** (0.06)	-0.20** (0.06)
Clustering level(s)	County	Geographic group	State	County by year
R ²	0.86	0.86	0.86	0.86
Observations	4,495,000	4,495,000	4,495,000	4,495,000
F-statistic	87.36	87.36	87.36	87.36
Dependent Variable Mean	69.85	69.85	69.85	69.85

Notes: This table reports the IV estimates using Equation (4) and Equation (5) when the standard errors are clustered at the different levels. The dependent variable is visits per thousand people on the day of exposure. All regressions include county-by-month, county-by-year, day-of-week, and month-by-year fixed effects, as well as flexible weather controls. Standard errors, clustered at the level(s) indicated in each column, are reported in parentheses. Geographic groups are shown in Figure A2. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Appendix Table A9. Robustness to Including Different Forms of Outcome

	Visits (per 1,000 people)	log(visit rate) $\times 100$	IHS(visits) $\times 100$
	(1)	(2)	(3)
PM2.5 ($\mu g/m^3$)	-0.20*** (0.01)	-0.38*** (0.02)	-0.38*** (0.02)
First-stage F stat	87.36	87.36	87.36
Dependent Variable Mean	69.85	69.85	69.85
Fixed Effects	Yes	Yes	Yes
R ²	0.86	0.89	0.99
Observations	4,495,000	4,494,488	4,495,000

Notes: This table reports OLS and IV estimates based on Equation (4) and Equation (5), with the dependent variable transformed using either the log of visit rates or the inverse hyperbolic sine (IHS) of visits. The main specification uses visits per thousand people on the day of exposure as the outcome. All regressions include county-by-month, county-by-year, day-of-week, and month-by-year fixed effects, as well as flexible weather controls. Standard errors clustered at the county level are reported in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Appendix Table A10. Robustness to COVID-19 Pandemic

	(1) Before COVID	(2) During COVID	(3) After COVID
PM2.5 ($\mu\text{g}/\text{m}^3$)	-0.12*** (0.02)	-0.10*** (0.01)	-0.10*** (0.03)
First-stage F stat	109.28	38.26	18.96
Dependent Variable Mean	64.97	65.79	83.26
Fixed Effects	Yes	Yes	Yes
R ²	0.88	0.90	0.89
Observations	2,464,500	837,000	1,156,300

Notes: This table presents IV estimates using Equation (4) and Equation (5) for different time periods. Before COVID denotes observations before March 15, 2020, when many U.S. states began implementing stay-at-home orders. During COVID denotes observations from March 15, 2020, to December 10, 2020, before the FDA issued an emergency use authorization for the COVID-19 vaccine. After COVID denotes observations from December 11, 2020 onward. The dependent variable is visits per thousand people on the day of exposure. All regressions include county-by-month, county-by-year, day-of-week, and month-by-year fixed effects, as well as flexible weather controls. Standard errors clustered at the county level are reported in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Appendix Table A11. Robustness to Excluding Counties without Satellite Data

	(1) With IDW log(visit rates) $\times$ 100	(2) Without IDW log(visit rates) $\times$ 100
PM2.5 ($\mu\text{g}/\text{m}^3$)	-0.20*** (0.01)	-0.17*** (0.02)
First-stage F stat	87.36	17.86
Dependent Variable Mean	69.85	66.04
Fixed Effects	Yes	Yes
R ²	0.88	0.88
Observations	4,495,000	1,684,900

Notes: This table reports the effect of daily PM2.5 on daily activities using only counties with satellite data. Column (1) presents my baseline specification, where counties without satellite data are interpolated using IDW. The dependent variable is visits per thousand people on the day of exposure. All regressions include county-by-month, county-by-year, day-of-week, and month-by-year fixed effects, as well as flexible weather controls. Standard errors clustered at the county level are reported in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.